DISTANCE ALGEBRAS OF FINITE DISTRIBUTIVE LATTICES ARE GORENSTEIN

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ABSTRACT. Let L be a finite distributive lattice and let Σ_L be the distance algebra of its Hasse graph. For every indecomposable injective Σ_L-module, we construct a projective resolution and use algebraic Morse theory to obtain a minimal resolution whose multiplicities are reduced Betti numbers of strict upset order complexes. As a consequence, Σ_L is (Iwanaga–)Gorenstein, with both self-injective dimensions bounded by the diameter of the Hasse graph of L. We further prove that the first two terms of each minimal resolution are multiplicity-free, whereas arbitrary multiplicities occur in every higher projective degree. We show in the appendix that the non-minimal resolution constructed is actually a special case of simplicial bar construction that turns out to be exact everywhere.

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1. INTRODUCTION

Magnitude homology was introduced by Hepworth and Willerton as a categorification of the magnitude of a graph [HW17]. Its connection with homological algebra was made explicit by Asao and Ivanov, who realized magnitude homology as a derived functor over the distance algebra [AI24]. Asao and Wakatsuki subsequently constructed minimal projective resolutions for the distance algebras of geodetic metric spaces [AW26]. In this paper we study the distance algebras associated with the Hasse graphs of finite distributive lattices, a family in which shortest paths are generally not unique.

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In our companion paper [CX26], we studies broader representation-theoretic and homological properties of distance algebras of finite graphs. In particular, it shows that the distance algebra of a finite modular lattice is Auslander–Gorenstein precisely when the lattice is a finite direct product of chains. For a distributive lattice $L = \mathcal{J}(P)$, this is equivalent to every connected component of P being a chain. The present paper establishes the Gorenstein and Iwanaga–Gorenstein properties for distance algebras of arbitrary finite distributive lattices under the corresponding assumptions on the coefficient ring.

By Birkhoff's fundamental theorem of distributive lattices, every finite distributive lattice has the form $\mathcal{J}(P)$ for a finite poset P. For the distance algebra $\Sigma_{\mathcal{J}(P)}$ over a commutative noetherian ring R, we construct a projective resolution of each module $I_X = D(\Sigma_{\mathcal{J}(P)} e_X)$, indexed by strict chains in a symmetric-difference poset. These resolutions have length at most $|P| - 1$. Together with the Gorenstein criterion of Iyengar–Krause [IK22, Theorem 4.6], this gives our first main result.

Theorem A. (see Corollary 3.4 and Proposition 3.12) *Let R be a commutative noetherian ring and let P be a non-empty finite poset. Let $\mathcal{J}(P)$ be the lattice of order ideals of P, and let $\Sigma_{\mathcal{J}(P)}$ be the distance R-algebra of its Hasse graph.*

1. *If R is Gorenstein, then $\Sigma_{\mathcal{J}(P)}$ is a Gorenstein R-algebra.*
2. *If R is Iwanaga–Gorenstein, then $\Sigma_{\mathcal{J}(P)}$ is Iwanaga–Gorenstein and*
$$\operatorname{idim}(\Sigma_{\mathcal{J}(P)})_{\Sigma_{\mathcal{J}(P)}} = \operatorname{idim}{}_{\Sigma_{\mathcal{J}(P)}}\Sigma_{\mathcal{J}(P)} \leq |P| - 1 + \operatorname{idim}_R R.$$
3. *If R is Iwanaga–Gorenstein and $|P| \geq 3$, then*
$$\operatorname{idim}(\Sigma_{\mathcal{J}(P)})_{\Sigma_{\mathcal{J}(P)}} = \operatorname{idim}{}_{\Sigma_{\mathcal{J}(P)}}\Sigma_{\mathcal{J}(P)} \leq |P| - 2 + \operatorname{idim}_R R.$$

Over a field $R = \Bbbk$, we use algebraic Morse theory [CLZ26] to obtain minimal projective resolutions of the indecomposable injectives. Their multiplicities and lengths are given by the reduced homology of the strict upset order complexes from Subsection 2.6; see Theorem 4.4 and Corollary 4.6. This also computes the self-injective dimension of Σ_L.

The combinatorics of order ideals forces multiplicity-freeness in the first two terms. For higher degrees, height-two posets identify the relevant order complexes with barycentric subdivisions of independence complexes of bipartite graphs. A realization theorem of Nagel and Reiner [NR09] then produces arbitrary multiplicities.

Theorem B. (see Theorem 4.4, Proposition 5.1, and Corollary 5.6) *For every finite poset P and every $X \in \mathcal{J}(P)$, we construct a minimal projective resolution of the indecomposable injective corresponding to X. These resolutions have the following properties.*

1. *The first two terms are multiplicity-free.*
2. *Multiplicities can be arbitrarily large in every higher projective degree.*

Organization of the paper. Section 2 recalls distance algebras, Gorenstein algebras, distributive lattices, and order complexes of strict upsets. Section 3 constructs the strict-chain projective resolutions, proves their exactness, and establishes Theorem A. Section 4 uses algebraic Morse theory to construct the minimal resolutions and compute their lengths. Sections 5 proves Theorem B. Appendix A places the strict chain construction in a more general framework for thin modules over arbitrary finite-dimensional algebras.

Conventions and notation. Throughout, R denotes a commutative noetherian ring with identity, and $\Bbbk$ denotes a field. All algebras are associative and unital. By an R-algebra we mean one that is finitely generated and projective as an R-module. Unless otherwise stated, modules are finitely generated right modules that are free over R. Unadorned tensor $\otimes$ between modules are tensor $\otimes_R$ over the ground ring R. All graphs, posets, lattices, and simplicial complexes are finite. The symbols idim and pdim denote injective and projective dimension, respectively. We write $D := \operatorname{Hom}_R(-, R)$. For a collection $\mathcal{X}$ of objects in an additive category $\mathcal{C}$, we write $\operatorname{add}\mathcal{X}$ for the additive closure of $\mathcal{X}$, that is, the full subcategory consisting of finite direct sum of direct summands of objects in $\mathcal{X}$.

Use of AI. The candidate complex is suggested to us by ChatGPT model 5.6 Sol, after we calculated the projective cover of the first syzygy (see subsection 5.1); the appendix records our

retrospect explanation in why one should consider such construction of complex. ChatGPT also suggested the strategy to show is exactness, helped us to refine the Morse matching described in Subsection 4.2, the entire subsection 5.2, and some of the figures throughout this notes. All mathematical arguments were independently verified and finalized by the authors. The authors take full responsibility for the accuracy, originality, and integrity of the paper.

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2. Preliminaries

2.1. **Duality and Gorenstein algebras.** Let A be a finite projective R-algebra.

Recall that a ring B is *Iwanaga–Gorenstein* if it is noetherian on both sides and

$$\operatorname{idim} B_B < \infty \qquad \text{and} \qquad \operatorname{idim}{}_B B < \infty.$$

A finite projective R-algebra A is called a *Gorenstein R-algebra* if $A_{\mathfrak{p}}$ is Iwanaga–Gorenstein for every $\mathfrak{p} \in \operatorname{Spec} R$ such that $A_{\mathfrak{p}} \neq 0$; see [IK22, Definition 1.1] and [GIK23, Definition 6.1]. When $R = \Bbbk$, this is precisely the Iwanaga–Gorenstein property. Following Buchweitz [Buc21, Section 7.6], the *dualizing bimodule* of A is

$$D(A) := \operatorname{Hom}_R(A, R) \text{ with action } (afb)(c) := f(bca) \quad (a, b, c \in A,\ f \in D(A)).$$

When R is Gorenstein, a criterion of Iyengar–Krause [IK22, Theorem 4.6] (see also [GIK23, Theorem 6.7]) states that A is a Gorenstein R-algebra if and only if

$$\operatorname{pdim}_A D(A) < \infty \qquad \text{and} \qquad \operatorname{pdim}_{A^{\mathrm{op}}} D(A) < \infty; \tag{2.1.1}$$

equivalently, $D(A)$ admits a bounded resolution by finitely generated projective modules on each side. For the distance algebras of finite distributive lattices, we will prove condition (2.1.1) without assuming that R is Gorenstein. When R is Gorenstein, the above criterion then shows that these are Gorenstein R-algebras.

An *A-lattice* is a right A-module whose underlying R-module is finitely generated and projective. We write $\operatorname{lat}(A)$ for the category of A-lattices, equipped with the exact structure inherited from the category of A-modules. Every short exact sequence in this category splits over R.

For finite projective algebras over a commutative noetherian ring, we recall the following fact from the theory of lattices over orders; see, for example, [CR81, Section 25]. The following formulation can be found in [BBI+25, Section 2]; we include a proof as it is straightforward.

Lemma 2.1. *The functor D induces an exact duality*

$$D : \operatorname{lat}(A)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{lat}(A^{\mathrm{op}}).$$

The category $\operatorname{lat}(A)$ *has enough projective and injective objects, given by* $\operatorname{add} A$ *and* $\operatorname{add} D(A)$*, respectively. Moreover, the projective dimension of an A-lattice in* $\operatorname{lat}(A)$ *equals its ordinary projective dimension.*

Proof. The functor D is exact on R-split short exact sequences, and evaluation gives a natural isomorphism $M \xrightarrow{\sim} D^2(M)$ for every $M \in \operatorname{lat}(A)$. This proves the duality assertion.

For $M \in \operatorname{lat}(A)$, choose a surjection $A^{\oplus m} \twoheadrightarrow M$ with m finite. It splits over R, so its kernel is again an A-lattice. Thus $\operatorname{lat}(A)$ has enough projectives, and its projective objects are precisely $\operatorname{add} A$. Assertion about enough injective follows by applying D to left projective modules. Since syzygy of $M \in \operatorname{lat}(A)$ stays in $\operatorname{lat}(A)$, projective resolution in $\operatorname{lat}(A)$ is the same as ordinary projective resolution; this last assertion follows. □

We call the objects of $\operatorname{add} D(A)$ *relative injective* A-modules. For $M \in \operatorname{lat}(A)$, write $\operatorname{idim}^{\mathrm{lat}}_A M$ for its injective dimension in $\operatorname{lat}(A)$, namely, the least length of a coresolution by the relative injectives, or ∞ if no finite such coresolution exists. When $R = \Bbbk$, relative injectives are the finitely generated ordinary injectives, and relative injective dimension agrees with ordinary injective dimension. We say that A is *self-injective* if $A \cong D(A)$ as right A-module; note that in this case they are also isomorphic as left A-module and so A is Iwanaga–Gorenstein.

Corollary 2.2. *Let A be a finite projective R-algebra. Then*

$$\operatorname{idim}_A^{\mathrm{lat}} A = \operatorname{pdim}_{A^{\mathrm{op}}} D(A), \qquad \operatorname{idim}_{A^{\mathrm{op}}}^{\mathrm{lat}} A = \operatorname{pdim}_A D(A).$$

If $\operatorname{idim}_R R < \infty$*, then*

$$\operatorname{idim} A_A \le \operatorname{idim}_A^{\mathrm{lat}} A + \operatorname{idim}_R R = \operatorname{pdim}_{A^{\mathrm{op}}} D(A) + \operatorname{idim}_R R,$$
$$\operatorname{idim} {}_A A \le \operatorname{idim}_{A^{\mathrm{op}}}^{\mathrm{lat}} A + \operatorname{idim}_R R = \operatorname{pdim}_A D(A) + \operatorname{idim}_R R.$$

In particular, if $\operatorname{idim}_R R$*,* $\operatorname{pdim}_{A^{\mathrm{op}}} D(A)$*, and* $\operatorname{pdim}_A D(A)$ *are all finite, then A is Iwanaga–Gorenstein. When $R = \Bbbk$ is a field, both inequalities are equalities.*

Remark 2.3. For related formulas for self-injective dimensions, see [GN01, Proposition 1.1(3)] and [IKWY15, Lemma 4.17]. We include an elementary proof of the above estimate for completeness.

Proof. Since A is finitely generated over the noetherian ring R, it is noetherian on both sides. By Lemma 2.1, the exact duality $D : \operatorname{lat}(A)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{lat}(A^{\mathrm{op}})$ swaps relative injective coresolutions of A_A with projective resolutions of ${}_A D(A)$. Since projective dimension in the lattice category agrees with ordinary projective dimension, we obtain $\operatorname{idim}_A^{\mathrm{lat}} A = \operatorname{pdim}_{A^{\mathrm{op}}} D(A)$. Applying the same argument to A^{op} gives the other relative dimension equality.

Assume now that $\operatorname{idim}_R R < \infty$. We claim that $\operatorname{idim} D(A)_A \le \operatorname{idim}_R R$. Indeed, for any right A-module N, so long as A is R-projective, any A-projective resolution of N is also an R-projective resolution. So tensor-Hom adjunction $\operatorname{Hom}_A(-, D(A)_A) \cong \operatorname{Hom}_R(-, R)$ gives

$$\operatorname{Ext}_A^q(N, D(A)_A) \cong \operatorname{Ext}_R^q(N, R) \qquad (q \ge 0),$$

and the claim follows.

Put $n := \operatorname{pdim}_{A^{\mathrm{op}}} D(A)$. The bound for $\operatorname{idim} A_A$ is immediate if $n = \infty$. If $n < \infty$, choose a resolution by finitely generated projective left A-modules

$$0 \longrightarrow Q_n \longrightarrow \cdots \longrightarrow Q_0 \longrightarrow {}_A D(A) \longrightarrow 0.$$

The modules Q_i and $D(A)$ are projective over R. Starting with the surjection $Q_0 \to D(A)$, it follows inductively that every syzygy is projective over R and every successive short exact sequence splits over R. Applying D and using $D^2(A) \cong A$ then yields an exact sequence of right A-modules

$$0 \longrightarrow A_A \longrightarrow D(Q_0) \longrightarrow \cdots \longrightarrow D(Q_n) \longrightarrow 0.$$

For every $0 \le i \le n$, we have $D(Q_i) \in \operatorname{add} D(A)_A$, and hence $\operatorname{idim}_A D(Q_i) \le \operatorname{idim}_R R$ by the claim in the previous paragraph. Dimension shifting along this exact sequence yields

$$\operatorname{idim} A_A \le \max_{0 \le i \le n} \{\operatorname{idim}_A D(Q_i) + i\} \le \operatorname{idim}_R R + n.$$

Applying the same argument to A^{op} gives $\operatorname{idim} {}_A A \le \operatorname{idim}_R R + \operatorname{pdim}_A D(A)$. Thus, if both $\operatorname{pdim}_{A^{\mathrm{op}}} D(A)$ and $\operatorname{pdim}_A D(A)$ are finite, A is Iwanaga–Gorenstein.

Finally, when $R = \Bbbk$, we have $\operatorname{idim}_R R = 0$, and relative injective dimension agrees with ordinary injective dimension. Putting this into the inequalities and the claim immediately follows. □

2.2. **Distance algebra.** Let $G = (V, E)$ be a connected combinatorial graph, where V is the set of vertices and E is the set of edges. In this paper, we assume that G contains neither loops nor multiple edges.

For vertices $x, y \in V$, we write $\{x, y\} \in E$ if x and y are connected by an edge; in this case, they are called *adjacent*. The *distance* between x and y, denoted by

$$d(x, y) = d_G(x, y),$$

is the length of a shortest path from x to y on G, and the *diameter* of G is the maximal distance in G, denoted by

$$\operatorname{diam}(G) := \max_{x, y \in V} d(x, y).$$

Definition 2.4. Let $G = (V, E)$ be a graph (without multi-edge and loops), and Q_G be its *double quiver*, i.e. the set of vertices of Q_G is V and there is a pair of arrows

$$x \underset{a_{yx}}{\overset{a_{xy}}{\rightleftarrows}} y \quad \text{for each edge } \{x, y\} \in E.$$

The *distance R-algebra* of G is defined to be $\Sigma_G := RQ_G/I_G$, where I_G is the ideal generated by the following relations:

(1) $p = 0$, for every path p such that $\ell(p) > d(s(p), t(p))$;
(2) $p = q$, for every pair of paths p, q in Q_G with

$$s(p) = s(q), \qquad t(p) = t(q), \qquad \ell(p) = d(s(p), t(p)) = \ell(q).$$

By definition, in Σ_G, a path is non-zero if and only if it is represented by a shortest path in G, and any two shortest paths with the same starting and ending vertices are identified. Hence, for any two vertices $x, y \in V$, there is a unique non-zero path from x to y modulo I_G. In particular, $e_x\Sigma_G e_y$ is a free R-module of rank one for all $x, y \in V$, and Σ_G is free of rank $|V|^2$ over R. When $R = \Bbbk$, the algebra Σ_G is Schurian.

We take an R-basis $\{b_{xy} \mid x, y \in V\}$ for Σ_G, where b_{xy} denotes the equivalence class of any shortest path from x to y on G, with the convention that $b_{xx} = e_x$ is the vertex idempotent associated to x. Then multiplication in Σ_G is given by

$$b_{xy}b_{yz} = \begin{cases} b_{xz}, & \text{if } d(x, z) = d(x, y) + d(y, z), \\ 0, & \text{otherwise.} \end{cases}$$

Note that b_{xy} is denoted by (x, y) and by e_{xy} in [AI24] and [AW26] respectively. Thus, each Σ_G is an R-algebra. Note that path length induces a natural grading on Σ_G; this matches the canonical (internal $\mathbb{R}$-)grading of magnitude homology.

The graphs that we concern in this note are the Hasse diagrams of finite distributive lattices L. We will simplify the notation and write Σ_L.

For each $x \in V(G)$, set

$$P_x := e_x\Sigma_G, \qquad I_x := D(\Sigma_G e_x).$$

These are projective and relative injective right Σ_G-modules, respectively, and

$$(\Sigma_G)_{\Sigma_G} = \bigoplus_{x \in V(G)} P_x, \qquad D(\Sigma_G)_{\Sigma_G} = \bigoplus_{x \in V(G)} I_x.$$

When $R = \Bbbk$, P_x and I_x are indecomposable, and I_x is the injective envelope of $S_x := P_x/\operatorname{rad} P_x$. Moreover, every finitely generated projective right Σ_G-module Q admits a decomposition $Q \cong \bigoplus_{x \in V(G)} P_x^{\oplus m_x}$, where the non-negative integers m_x are uniquely determined by the Krull–Schmidt theorem. We call m_x the *multiplicity* of P_x in Q and write $[Q : P_x] := m_x$. We call Q *multiplicity-free* if $[Q : P_x] \leq 1$ for every $x \in V(G)$.

Path reversal defines an R-linear anti-involution

$$\tau : \Sigma_G \longrightarrow \Sigma_G, \qquad b_{xy} \longmapsto b_{yx}.$$

Thus τ induces an isomorphism of R-algebras $\Sigma_G^{\mathrm{op}} \cong \Sigma_G$.

For a right Σ_G-module M, let $\tau_* M$ be the left Σ_G-module with action $a \cdot m := m\tau(a)$. This gives a covariant exact equivalence between right and left module categories, preserving projectives and injectives. Its inverse is defined by the same construction for left modules. Since $\tau(e_x) = e_x$, there is an isomorphism of left Σ_G-modules

$$\tau_* I_x \xrightarrow{\sim} D(e_x\Sigma_G), \qquad f \longmapsto \big(b \longmapsto f(\tau(b))\big). \tag{2.2.1}$$

Thus, we have the following lattice version of duality on distance algebras [CX26, Section 2].

Proposition 2.5. *For $M \in \operatorname{lat}(\Sigma_G)$, equip $D(M)$ with the right Σ_G-action*

$$(f \cdot a)(m) := f\big(m\tau(a)\big).$$

With this right action, D defines an exact contravariant self-duality of $\operatorname{lat}(\Sigma_G)$, with $D(P_x) \cong I_x$ and $D(I_x) \cong P_x$. Consequently, for every $M \in \operatorname{lat}(\Sigma_G)$,

$$\operatorname{idim}^{\operatorname{lat}}_{\Sigma_G} M = \operatorname{pdim}_{\Sigma_G} D(M), \qquad \operatorname{pdim}_{\Sigma_G} M = \operatorname{idim}^{\operatorname{lat}}_{\Sigma_G} D(M).$$

Proof. Compose the duality of Lemma 2.1 with the equivalence induced by τ. The asserted action and the isomorphisms follow from $\tau^2 = \mathrm{id}$ and (2.2.1). This self-duality exchanges projective resolutions and relative injective coresolutions, giving the dimension equalities. □

Applying Corollary 2.2 to Σ_G gives the following criterion. Here $D(\Sigma_G)$ is regarded as the dualizing bimodule with its natural left and right actions.

Corollary 2.6. *Let* $n := \max_{x\in V(G)} \operatorname{pdim}_{\Sigma_G} I_x \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$. *Then*

$$\operatorname{pdim}_{\Sigma_G} D(\Sigma_G) = \operatorname{pdim}_{\Sigma_G^{\mathrm{op}}} D(\Sigma_G) = \operatorname{idim}^{\mathrm{lat}}_{\Sigma_G} \Sigma_G = n. \tag{2.2.2}$$

(1) *If* R *is Gorenstein, then* Σ_G *is a Gorenstein* R*-algebra if and only if* $n < \infty$.
(2) *If* R *is Iwanaga–Gorenstein and* $n < \infty$, *then* Σ_G *is Iwanaga–Gorenstein and*

$$\operatorname{idim}(\Sigma_G)_{\Sigma_G} = \operatorname{idim}{}_{\Sigma_G}\Sigma_G \leq n + \operatorname{idim}_R R.$$

(3) *When* $R = \Bbbk$ *is a field, both self-injective dimensions equal* n.

Proof. The decomposition $D(\Sigma_G)_{\Sigma_G} = \bigoplus_{x\in V(G)} I_x$ gives $\operatorname{pdim}_{\Sigma_G} D(\Sigma_G) = n$. Summing (2.2.1) over the vertices gives an isomorphism of left Σ_G-modules

$$\tau_*\big(D(\Sigma_G)_{\Sigma_G}\big) \cong {}_{\Sigma_G}D(\Sigma_G).$$

Since τ_* preserves projective dimensions, we also have $\operatorname{pdim}_{\Sigma_G^{\mathrm{op}}} D(\Sigma_G) = n$. The remaining assertions follow immediately from Corollary 2.2, the Gorenstein criterion (2.1.1), and the equivalence induced by τ. □

2.3. **Simplicial complexes, homologies, and order complexes.** Unless a coefficient ring is specified, all chain groups and homology groups are taken over R.

A finite *simplicial complex* Δ on a finite vertex set V is a collection of subsets of V such that $\sigma \in \Delta$ and $\tau \subseteq \sigma$ imply $\tau \in \Delta$. The elements of Δ are called its *faces*, and a face σ has *dimension* $|\sigma| - 1$. We denote by Δ_r the set of r-dimensional faces of Δ. We allow the void complex $\emptyset$, which has no faces. We fix a total order on the vertex set when necessary, then an r-dimensional face equipped with the the inherited ordering is called an *r-chain* in Δ.

For a finite poset $(P, <)$, its *order complex* $\Delta(P)$ is the simplicial complex whose r-chains are the strict chains $x_0 < x_1 < \cdots < x_r$ in P. In particular, $\Delta(\emptyset) = \{\emptyset\}$ is not the void complex $\emptyset$. Note that we do not need to linearize P to a total order since every r-simplex is already totally ordered by definition.

For $r \geq 0$, let $C_r(\Delta) := R\Delta_r$ be the free R-module generated by the r-chains of Δ. We denote the corresponding basis element by $\langle x_0 < x_1 < \cdots < x_r\rangle$.

For $r \geq 1$, the simplicial *boundary map* is given by

$$\partial_r\langle x_0 < \cdots < x_r\rangle := \sum_{i=0}^{r} (-1)^i \langle x_0 < \cdots < \widehat{x_i} < \cdots < x_r\rangle,$$

where $\widehat{x_i}$ means that x_i is removed from the chain. When we want to write in a more compact form, for a chain $x_* = (x_0 < x_1 < \cdots < x_r)$, we write $x_*^{\widehat{i}}$ for the chain obtained from x_* by deleting x_i; thus, we have $\partial_r\langle x_*\rangle = \sum_{i=0}^{r}(-1)^i\langle x_*^{\widehat{i}}\rangle$. The boundary maps satisfy $\partial_{r-1}\partial_r = 0$. The *ordinary simplicial homology* of Δ is the homology of the chain complex

$$C_\bullet(\Delta)\colon \quad \cdots \longrightarrow C_2(\Delta) \xrightarrow{\partial_2} C_1(\Delta) \xrightarrow{\partial_1} C_0(\Delta) \longrightarrow 0,$$

and is denoted by $H_r(\Delta) := \operatorname{Ker}\partial_r / \operatorname{Im}\partial_{r+1}$.

Set $C_{-1}(\Delta) := R$ and define the *augmentation map*

$$\partial_0\colon C_0(\Delta) \longrightarrow C_{-1}(\Delta), \qquad \partial_0\langle x\rangle := 1.$$

Adjoining to $C_\bullet(\Delta)$ yields a chain complex starting in degree -1:

$$\widetilde{C}_\bullet(\Delta)\colon \quad \cdots \longrightarrow C_1(\Delta) \xrightarrow{\partial_1} C_0(\Delta) \xrightarrow{\partial_0} C_{-1}(\Delta) \longrightarrow 0. \tag{2.3.1}$$

Its homology is the *reduced homology* $\widetilde{H}_r(\Delta)$. In particular, we have

$$\widetilde{H}_{-1}(\emptyset) = R, \qquad \text{and} \qquad \widetilde{H}_r(\Delta) = H_r(\Delta) \quad \text{for } r > 0.$$

For later use, we call $C_{-1}(\Delta)$ the *augmentation part* and the non-negative degree part $C_\bullet(\Delta)$ the *resolution part* of $\widetilde{C}_\bullet(\Delta)$. If clarification of ground ring is needed, we will add '$;R$' inside the brackets, for example, $C_\bullet(\Delta; R)$ for the R-valued chain complex.

2.4. **More on order complexes.** A simplicial complex K is a *cone with vertex p* if $p \in V(K)$ and $\sigma \cup \{p\} \in K$ for every $\sigma \in K$. A cone has zero reduced homology over R. To see this, order its vertices with p first and define

$$h_{-1} : R \longrightarrow C_0(K), \qquad 1 \longmapsto \langle p \rangle,$$

and, for $k \geq 0$, define

$$h_k : C_k(K) \longrightarrow C_{k+1}(K), \qquad \langle v_0 < \cdots < v_k \rangle \longmapsto \begin{cases} \langle p < v_0 < \cdots < v_k \rangle, & p \notin \{v_0, \ldots, v_k\}, \\ 0, & p \in \{v_0, \ldots, v_k\}. \end{cases}$$

These maps satisfy $\partial_0 h_{-1} = \mathrm{id}_R$ and $\partial_{k+1} h_k + h_{k-1} \partial_k = \mathrm{id}_{C_k(K)}$ for $k \geq 0$, so they contract the augmented complex $\widetilde{C}_\bullet(K)$. In particular, the order complex of a non-empty poset with a least or greatest element is a cone.

Definition 2.7. Let K be a finite simplicial complex on a vertex set V. For a face $F \in K$, define its *link* of F in K by

$$\mathrm{lk}_K F := \{G \subseteq V \setminus F : F \cup G \in K\}.$$

Define the *face poset* and the *non-empty-face poset* of K as

$$\mathcal{F}(K) := \{\text{faces of } K\}, \quad \mathcal{F}^\times(K) := \mathcal{F}(K) \setminus \{\emptyset\},$$

respectively, ordered by inclusion. The *barycentric subdivision* $\mathrm{sd}\, K$ of K is the order complex of the non-empty-face poset of K.

For a finite simplicial complex K, write $|K|$ for its geometric realization. We identify simplicial homology with the singular homology of the realization. Barycentric subdivision preserves the geometric realization, and hence preserves reduced homology over R.

For a finite set E with $|E| \geq 2$, the order complex of the power set 2^E, whose partial order given by set inclusion, is a $(|E|-1)$-simplex on E. Its boundary is $\partial(2^E) = 2^E \setminus \{E\}$, so

$$\Delta\big(2^E \setminus \{\emptyset, E\}\big) = \Delta\big(\mathcal{F}^\times(\partial(2^E))\big) = \mathrm{sd}(\partial(2^E)), \quad \text{and} \quad |\mathrm{sd}(\partial(2^E))| \cong S^{|E|-2}. \tag{2.4.1}$$

Thus omitting E passes to the boundary of the simplex, and omitting $\emptyset$ passes to its non-empty-face poset.

Lemma 2.8. *Let $\mathcal{F}(K)_{>F} := \{F' \in K \mid F \subsetneq F'\}$. Then there is an isomorphism of simplicial complexes*

$$\Delta\big(\mathcal{F}(K)_{>F}\big) \cong \mathrm{sd}(\mathrm{lk}_K F).$$

In particular, we have $\widetilde{H}_k(\mathrm{lk}_K F) \cong \widetilde{H}_k(\Delta(\mathcal{F}(K)_{>F}))$ for all $k \geq -1$.

Proof. The first part is known in [Mun84, Lemma 15.3]; we can also obtain it using the poset isomorphism

$$\mathcal{F}(K)_{>F} := \{F' \in K \mid F \subsetneq F'\} \to \mathcal{F}^\times(\mathrm{lk}_K F), \qquad F' \mapsto F' \setminus F, \text{ with inverse } F \cup H \hookleftarrow H.$$

The second part follows from

$$\widetilde{H}_k(\mathrm{lk}_K F) \cong \widetilde{H}_k\big(\mathrm{sd}(\mathrm{lk}_K F)\big) \cong \widetilde{H}_k\big(\Delta(\mathcal{F}(K)_{>F})\big),$$

where the first isomorphism uses the fact that barycentric subdivision preserves reduced homology [Mun84, Theorem 17.2]. □

We also recall an order homotopy theorem for maps between order complexes.

Let $f : Q \to Q'$ be an order-preserving map between two finite posets Q, Q'. Define, for all $k \geq 0$, a map $f_{\#,k} : C_k(\Delta(Q)) \to C_k(\Delta(Q'))$ given by

$$\langle x_0 < \cdots < x_k \rangle \mapsto \begin{cases} \langle f(x_0) < \cdots < f(x_k) \rangle, & f(x_0), \ldots, f(x_k) \text{ are pairwise distinct}, \\ 0, & \text{otherwise}. \end{cases}$$

We also set $f_{\#,-1} = \mathrm{id}_R$. Then $f_\#$ is a chain map on the chain complexes and their augmentations.

Lemma 2.9. *Let $f, g : Q \to Q'$ be order-preserving maps between finite posets, and suppose that $f(q) \leq g(q)$ for every $q \in Q$. The induced chain maps $f_\#, g_\#$ are chain homotopic.*

Proof. This is the chain complex version of [Bjö95, Theorem 10.11]; see also [Mun84, Theorem 13.3]. We can verify it directly as follows. For $k \geq 0$, define a map $h_k : C_k(\Delta(Q)) \longrightarrow C_{k+1}(\Delta(Q'))$ by

$$\langle x_0 < \cdots < x_k \rangle \mapsto \sum_{j=0}^{k} (-1)^j \langle f(x_0) \leq \cdots \leq f(x_j) \leq g(x_j) \leq \cdots \leq g(x_k) \rangle.$$

Every chain displayed on the right is weakly increasing, so to get well-defined map we interpret each summand containing repeated members as zero. Set also $h_{-1} = 0$. Then it is routine to check that

$$\partial^{Q'}_{k+1} h_k + h_{k-1} \partial^{Q}_{k} = g_{\#,k} - f_{\#,k},$$

which gives the desired homotopy. □

The following consequence of Alexander duality will be useful.

Lemma 2.10. *Assume that $|P| \geq 3$, and let K be a proper subcomplex of $\Delta\big(2^P \setminus \{\emptyset, P\}\big)$. Then the boundary map $\partial_{|P|-2} : C_{|P|-2}(K) \longrightarrow C_{|P|-3}(K)$ has an R-linear left inverse.*

Proof. Put $d := |P| - 2 \geq 1$. If $K = \emptyset$ there is nothing to prove, so assume $K \neq \emptyset$. Since P is finite, each $C_q(K) = C_q(K; \mathbb{Z}) \otimes_{\mathbb{Z}} R$ is a finitely generated free R-module. Let

$$C^q(K; R) := \mathrm{Hom}_R\big(C_q(K), R\big), \qquad \delta^q := \mathrm{Hom}_R(\partial_{q+1}, R) \colon C^q(K; R) \to C^{q+1}(K; R)$$

be the cochain complex. Each $C^q(K; R) \cong C^q(K; \mathbb{Z}) \otimes_{\mathbb{Z}} R$ is again finitely generated free over R.

We first show that $\delta^{d-1} \colon C^{d-1}(K; \mathbb{Z}) \to C^d(K; \mathbb{Z})$ is surjective. Indeed, since $|\Delta(2^P \setminus \{\emptyset, P\})| \cong S^d$ by (2.4.1), K is a proper subcomplex, $|K|$ is a compact, locally contractible, non-empty, proper subspace of S^d. In particular, we have $S^d \setminus |K| \neq \emptyset$, and Alexander duality [Hat02, Cor. 3.45] gives

$$H^d(K; \mathbb{Z}) \;\cong\; \widetilde{H}^d(|K|; \mathbb{Z}) \;\cong\; \widetilde{H}_{-1}(S^d \setminus |K|; \mathbb{Z}) \;=\; 0,$$

where the first isomorphism uses $d \geq 1$, and the last equality holds because reduced homology in degree -1 vanishes for a non-empty space. As K has no simplices of dimension strictly larger than d, we have $C^{d+1}(K; \mathbb{Z}) = 0$, so $H^d(K; \mathbb{Z}) = \operatorname{coker} \delta^{d-1}$, which means that δ^{d-1} is surjective, as claimed.

Now we base change back to R. Since $- \otimes_{\mathbb{Z}} R$ is right exact, $\delta^{d-1} \colon C^{d-1}(K; R) \to C^d(K; R)$ is still surjective with a free (hence, projective) codomain, so there is an R-linear right inverse $s : C^d(K; R) \to C^{d-1}(K; R)$ of δ^{d-1}.

Now the claim follows by dualizing back to the chain groups. Namely, since the chain groups are finitely generated free over R, we have $C_q(K; R) \cong \mathrm{Hom}_R(C^q(K; R), R)$ and $\mathrm{Hom}_R(\delta^{d-1}, R) = \partial_d$. Applying $\mathrm{Hom}_R(-, R)$ to $\delta^{d-1} \circ s = \mathrm{id}$ thus yields the required $\mathrm{Hom}_R(s, R) \circ \partial_d = \mathrm{id}$. □

2.5. **Finite posets and distributive lattices.** Recall that a *lattice* is a poset L in which every pair $x, y \in L$ has a greatest lower bound, called the *meet* and denoted by $x \wedge y$, and a least upper bound, called the *join* and denoted by $x \vee y$. A lattice is *distributive* if

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z), \qquad x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$$

for all $x, y, z \in L$. A finite lattice has a least element 0 and a greatest element 1.

For a finite poset P, we denote by $\max(P)$ the set of its maximal elements. An *order ideal* of P is a subset $X \subseteq P$ such that if $y \in X$ and $x \leq y$ then $x \in X$. We denote by $\mathcal{J}(P)$ the set of order ideals of P, ordered by inclusion. It is a finite distributive lattice with $X \wedge Y = X \cap Y$ and $X \vee Y = X \cup Y$.

Conversely, Birkhoff's fundamental theorem of finite distributive lattices states that every finite distributive lattice is isomorphic to $\mathcal{J}(P)$ for some finite poset P; see, for example, [Grä03].

The *Hasse graph* of a finite poset Q is the undirected graph whose vertex set is Q, in which two elements are adjacent if one covers the other, i.e. an edge is given by $\{x, y\}$ with $x < y$ and there is no $z \in Q$ satisfying $x < z < y$. We write $d_Q(x, y)$, or simply $d(x, y)$, for the corresponding graph distance. Moreover, for subsets $X, Y \subseteq Q$, their *symmetric difference* is $X \triangle Y := (X \setminus Y) \sqcup (Y \setminus X)$.

Lemma 2.11. *Let X, Y, Z be subsets of a finite set. Then the following hold.*

(1) *$X \triangle Y = Y \triangle X$, $X \triangle X = \emptyset$, and $(X \triangle Y) \triangle Z = X \triangle (Y \triangle Z)$.*
(2) *$(X \triangle Y) \triangle (X \triangle Z) = Y \triangle Z$, $X \triangle (X \triangle Y) = Y$.*

(3) *Let* $\nabla_X^{Y,Z} := (X\cap Y)\cup(X\cap Z)\cup(Y\cap Z)$ *(note that the position of* X,Y,Z *around* ∇ *does not matter). Then* $(X\triangle Y)\cap(X\triangle Z) = \big(X\setminus(Y\cup Z)\big)\sqcup\big((Y\cap Z)\setminus X\big) = X\triangle\nabla_X^{Y,Z}$.

(4) $(X\triangle Y)\setminus(X\triangle Z) = \big(X\cap(Z\setminus Y)\big)\sqcup\big((Y\setminus Z)\setminus X\big)$.

(5) *If* $Y\subseteq Z$, *then* $(X\triangle Y)\subseteq(X\triangle Z) \iff X\cap(Z\setminus Y)=\emptyset$. *More generally, we have*

$$(X\triangle Y)\subseteq(X\triangle Z) \iff \begin{cases} X\cap(Z\setminus Y)=\emptyset,\\ Y\setminus Z\subseteq X.\end{cases}$$

(6) *If* $(X\triangle Z)\subseteq(X\triangle Y)$, *then*

$$\begin{aligned} X\triangle Y &= (X\triangle Z)\sqcup(Z\setminus Y)\sqcup(Y\setminus Z),\\ X\triangle(Y\cap Z) &= (X\triangle Z)\sqcup(Z\setminus Y),\\ X\triangle(Y\cup Z) &= (X\triangle Z)\sqcup(Y\setminus Z).\end{aligned}$$

(7) $|Y| = |X| + |X\triangle Y|$ *if and only if* $X\subseteq Y$.

Proof. (1), (2), (3) are elementary set theory exercise which we leave for the reader to check.

(4) Intersecting $(X\triangle Y)\setminus(X\triangle Z)$ with X yields the left set in the decomposition, and removing X yields the second.

(5) The 'more generally' part follows from (4) as left-hand side is equivalent to $(X\triangle Y)\setminus(X\triangle Z)=\emptyset$. The first part of the claim follows from this general case immediately.

(6) The hypothesis $(X\triangle Z)\subseteq(X\triangle Y)$ gives $Z\setminus Y\subseteq X$ and $(Y\setminus Z)\cap X=\emptyset$ by (5) (with Y and Z exchanged). Thus removing $Z\setminus Y$ from Z, or adjoining $Y\setminus Z$ to Z, adds precisely the corresponding set to $X\triangle Z$. The three identities then follow from

$$\begin{aligned} Y &= \big(Z\setminus(Z\setminus Y)\big)\sqcup(Y\setminus Z),\\ Y\cap Z &= Z\setminus(Z\setminus Y),\\ Y\cup Z &= Z\sqcup(Y\setminus Z).\end{aligned}$$

Note that the unions are disjoint because $Z\setminus Y$ and $Y\setminus Z$ are disjoint from $X\triangle Z$ and from each other.

(7) We always have $|X|+|X\triangle Y|-|Y| = 2|X\setminus Y|$, and so $|Y| = |X|+|X\triangle Y|$ holds precisely when $X\setminus Y=\emptyset$, or equivalently, $X\subseteq Y$. □

Lemma 2.12. *For any* $X,Y,Z\in\mathcal{J}(P)$, *one has*

$$d(X,Y) = |X\triangle Y|, \quad \text{and} \quad d(X,Y) = d(X,Z)+d(Z,Y) \Longleftrightarrow X\triangle Z\subseteq X\triangle Y.$$

Proof. Every edge in the Hasse graph of $\mathcal{J}(P)$ changes an order ideal by exactly one element. Hence every path from X to Y has length at least $|X\triangle Y|$. Conversely, one may first remove the elements of $X\setminus Y$, choosing a maximal remaining element at each step, and then add the elements of $Y\setminus X$, choosing a minimal missing element at each step. Every intermediate subset is an order ideal, and this gives a path of length $|X\triangle Y|$. By Lemma 2.11(2), $Z\triangle Y = (X\triangle Z)\triangle(X\triangle Y)$. Hence the second assertion follows from the distance formula and Lemma 2.11(7), applied to $X\triangle Z$ and $X\triangle Y$. □

Lemma 2.13 (cf. [CX26, Lemma 6.3 and Corollary 7.24(2)])**.** *For* $X\in\mathcal{J}(P)$, *the projective* $\Sigma_{\mathcal{J}(P)}$*-module* P_X *is relative injective if and only if* $P\setminus X\in\mathcal{J}(P)$. *In this case,* $P_X\cong I_{P\setminus X}$ *and* $P_{P\setminus X}\cong I_X$.

Proof. $\Rightarrow$: Suppose that P_X is relative injective, that is, $P_X\in\operatorname{add} D(\Sigma_{\mathcal{J}(P)})$. Choose a maximal ideal $\mathfrak{m}$ of R. Since $\Sigma_{\mathcal{J}(P)}$ is free over R of finite rank, we have

$$D(\Sigma_{\mathcal{J}(P)})\otimes_R R/\mathfrak{m}\cong \operatorname{Hom}_{R/\mathfrak{m}}\big(\Sigma_{\mathcal{J}(P)}\otimes_R R/\mathfrak{m}, R/\mathfrak{m}\big).$$

Thus $P_X\otimes_R R/\mathfrak{m}$ is injective over $\Sigma_{\mathcal{J}(P)}\otimes_R R/\mathfrak{m}$. By [CX26, Lemma 6.3 and Corollary 7.24(2)], we obtain $P\setminus X\in\mathcal{J}(P)$.

$\Leftarrow$: Suppose that $P\setminus X\in\mathcal{J}(P)$. By Lemma 2.12, for every $Y\in\mathcal{J}(P)$, we have $d(X,Y)+d(Y,P\setminus X) = |P| = d(X,P\setminus X)$, which then implies that $b_{XY}b_{Z,P\setminus X} = \delta_{Y,Z}b_{X,P\setminus X}$ for any $Y,Z\in\mathcal{J}(P)$. Define

$$\phi: P_X\longrightarrow I_{P\setminus X}, \qquad \phi(p)(q) := \text{the coefficient of } b_{X,P\setminus X} \text{ in } pq,$$

where $p \in P_X = e_X \Sigma_{\mathcal{J}(P)}$ and $q \in \Sigma_{\mathcal{J}(P)} e_{P\setminus X}$. It is routine to verify that this is a homomorphism of modules. The multiplication $b_{XY} b_{Z,\,P\setminus X} = \delta_{Y,Z} b_{X,\,P\setminus X}$ implies that ϕ sends the basis $\{b_{XY}\}_{Y\in\mathcal{J}(P)}$ to the dual R-basis of $\{b_{Y,\,P\setminus X}\}_{Y\in\mathcal{J}(P)}$. Thus $P_X \cong I_{P\setminus X}$ and is relative injective. Swapping X and $P \setminus X$ gives $P_{P\setminus X} \cong I_X$. □

Example 2.14. Up to isomorphism, there are exactly four posets P with $|P| \leq 2$. In the following, A_n denotes the path graph on n vertices.

(1) If $P = \emptyset$, then $\mathcal{J}(P) = \{\emptyset\}$ and $\Sigma_{\mathcal{J}(P)} \cong R$. Thus, Σ_P is self-injective ($\operatorname{idim}^{\mathrm{lat}} \Sigma_P = 0$).

(2) If $P = \{*\}$, then $\mathcal{J}(P) = \{\emptyset \subset (*)\}$ and $\Sigma_{\mathcal{J}(P)} \cong \Sigma_{A_2}$. The algebra Σ_{A_2} is given by the quiver

$$1 \underset{b_{21}}{\overset{b_{12}}{\rightleftarrows}} 2 \quad \text{with relations } b_{12}b_{21} = 0 = b_{21}b_{12}.$$

Then by Lemma 2.13 Σ_{A_2} is self-injective.

(3) If $P = \{a, b\}$ with a, b incomparable, then $\mathcal{J}(P)$ is the Boolean lattice of rank two, which is the direct product of two copies of $\{1 < 2\}$. The property $P \setminus X \in \mathcal{J}(P)$ for every $X \in \mathcal{J}(P)$ still holds, and so it follows from Lemma 2.13 that $\Sigma_{J(P)}$ is self-injective. Note that the Hasse graph of $J(P)$ is the square Q_2, which is the Cartesian product of two copies of A_2, and so it follows from [CX26, Proposition 3.2] that

$$\Sigma_{\mathcal{J}(P)} \cong \Sigma_{Q_2} \cong \Sigma_{A_2} \otimes_R \Sigma_{A_2},$$

and one can then deduce self-injective of $\Sigma_{\mathcal{J}(P)}$ by the fact that tensor product over the base ring R preserves self-injectivity.

(4) If $P = \{a < b\}$, then $\mathcal{J}(P) = \{\emptyset \subset (a) \subset (a < b)\} \cong \{1 < 2 < 3\}$ and $\Sigma_{\mathcal{J}(P)} \cong \Sigma_{A_3}$. The algebra Σ_{A_3} is given by the quiver

$$1 \underset{b_{21}}{\overset{b_{12}}{\rightleftarrows}} 2 \underset{b_{32}}{\overset{b_{23}}{\rightleftarrows}} 3$$

with every path of length two that returns to its starting vertex equal to zero.

By Lemma 2.13, $P_1 \cong I_3$ and $P_3 \cong I_1$, whereas P_2 is not relative injective. Let $\{f_i\}_{i=1}^3$ be the R-basis of $I_2 = D(\Sigma_{A_3} e_2)$ dual to $\{b_{i2}\}_{i=1}^3$. There is an exact sequence

$$0 \longrightarrow P_2 \xrightarrow{\iota} P_1 \oplus P_3 \xrightarrow{\pi} I_2 \longrightarrow 0,$$

where $\iota(p) = (b_{12}p, -b_{32}p)$ and $\pi(p, q) = f_1 p + f_3 q$. This sequence is a relative injective coresolution of P_2 of length one, and $\operatorname{idim}^{\mathrm{lat}} \Sigma_{A_3} = 1$. The same equality holds on the left by Corollaries 2.2 and 2.6. For the special case where R is a field, see [CX26, Proposition 7.6].

2.6. **Order complexes of strict upsets.** Let P be a finite poset, fix $X \in \mathcal{J}(P)$.

We write $\underline{Z} := X \triangle Z$ (and $\underline{Z}_X$ if clarification is needed). Consider the poset

$$\mathcal{D}_X := \{X \triangle Z \mid Z \in \mathcal{J}(P)\},$$

ordered by inclusion.

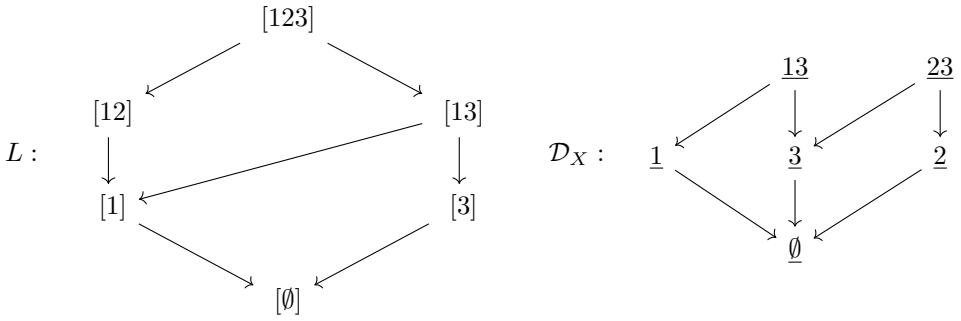
For $Z \in \mathcal{J}(P)$, let $\mathcal{D}_X^{>\underline{Z}}$ be the upset of $\mathcal{D}_X$ containing all elements strictly larger than Z; we call this *strict upset above* $\underline{Z}$ for convenience. Denote by

$$\Delta_X := \Delta(\mathcal{D}_X), \qquad \text{and} \qquad \Delta^{>\underline{Z}} := \Delta_X^{>\underline{Z}} := \Delta(\mathcal{D}_X^{>\underline{Z}})$$

the associated order complexes. Note that the vertices of $\Delta^{>\underline{Z}}$ are the elements $\underline{T} > \underline{Z}$, and its r-dimensional faces are the strict chains $\underline{T_0} < \underline{T_1} < \cdots < \underline{T_r}$, which means that it is a subcomplex of Δ_X.

Example 2.15. Let $P = \{1 < 2\} \sqcup \{3\}$. For more compact display, we use, for instance, $[ijk]$ to denote $\{i, j, k\} \subset P$, and write $\underline{xy}$ when $\underline{Z} = \{x, y\}$. We consider the case when $X = \{1\} = [1]$.

Z	$[1]$	$[\emptyset]$	$[12]$	$[13]$	$[3]$	$[123]$
$\underline{Z} = X \triangle Z$	$\underline{\emptyset}$	$\underline{1}$	$\underline{2}$	$\underline{3}$	$\underline{13}$	$\underline{23}$



Note that $\mathcal{D}_X$ is really just a re-orientation of L.

The six strict upsets $\mathcal{D}_X^{>\underline{Z}}$ and their order complexes $\Delta_X^{>\underline{Z}}$ are displayed below. In this example, every $\mathcal{D}_X^{>\underline{Z}}$ has height at most one, so each $\Delta_X^{>\underline{Z}}$ is at most 1-dimensional and may be visualized as the underlying graph of the Hasse diagram of $\Delta_X^{\underline{Z}}$.

Z	$\mathcal{D}_X^{>\underline{Z}}$	$\Delta_X^{>\underline{Z}}$
$[1]$	$\{\underline{1},\underline{2},\underline{3},\underline{13},\underline{23}\}$	$\underline{1}-\underline{13}-\underline{3}-\underline{23}-\underline{2}$
$[\emptyset]$	$\{\underline{13}\}$	$\{\underline{13}\}$
$[12]$	$\{\underline{23}\}$	$\{\underline{23}\}$
$[13]$	$\{\underline{13},\underline{23}\}$	$\{\underline{13}\}\sqcup\{\underline{23}\}$
$[3]$	$\emptyset$	$\{\underline{\emptyset}\}$
$[123]$	$\emptyset$	$\{\underline{\emptyset}\}$

Consequently,

$$\widetilde{H}_r(\Delta_{[1]}^{>\underline{Z}})=0 \quad \text{except for} \quad \widetilde{H}_0(\Delta_{[1]}^{>\underline{3}})\cong R(\langle\underline{13}\rangle-\langle\underline{23}\rangle), \quad \widetilde{H}_{-1}(\Delta_{[1]}^{>\underline{13}})\cong R, \quad \widetilde{H}_{-1}(\Delta_{[1]}^{>\underline{23}})\cong R.$$

3. A projective resolution from strict chains

We retain the notation and assumptions of Subsection 2.6. In this section we construct a complex $\mathcal{P}_\bullet$ of projective A-modules indexed by the strict chains in $\mathcal{D}_X$, equip it with an augmentation to I_X, and prove that the augmented complex is exact.

3.1. **The candidate complex.** To distinguish between different copies of the same module in a direct sum, we will use $P_Z^{(i)}$ to denote the copy of P_Z labeled by i in some indexing set. Similarly, for $p\in P_Z$, we write $p^{(i)}$ for the element p in the i-labelled copy of P_Z; this is nothing but a compact way to write $\iota_i(p)$ where ι_i is the canonical inclusion of $P_Z^{(i)}$ into the direct sum that contains it.

For $k\geq 0$, we write

$$\underline{Z}_*=(\underline{Z}_0<\cdots<\underline{Z}_k)\in\Delta(\mathcal{D}_X)_k$$

for a strict chain and it is understood that Z_i is the order ideal such that $Z_i\triangle X=\underline{Z}_i$. For $0\leq i\leq k$, define

$$\underline{Z}_*^{\widehat{i}}:=(\underline{Z}_0<\cdots<\widehat{\underline{Z}_i}<\cdots<\underline{Z}_k)$$

to be the strict chain obtained by deleting $\underline{Z}_i$. When $k=0$, the chain $\underline{Z}_*^{\widehat{0}}$ is empty.

For $k\geq 0$, define $\mathcal{P}_k$ to be the direct sum of projective A-module P_Z over all k-chains in $\Delta_{X,k}$. Observe that, when $k=0$ (so $\Delta_{X,0}=\mathcal{D}_X=\mathcal{J}(P)$), we have

$$\mathcal{P}_0=\bigoplus_{Z\in\mathcal{J}(P)}P_Z^{(\underline{Z})}\cong\bigoplus_{Z\in\mathcal{J}(P)}P_Z\otimes_R C_{-1}(\Delta^{>\underline{Z}})$$

via $p^{(\underline{Z})}\mapsto p\otimes 1\in P_Z\otimes_R C_{-1}(\Delta^{>\underline{Z}})$. On the other hand, when $k>0$, a chain $\underline{Z}_*=(\underline{Z}_0<\underline{Z}_1<\cdots<\underline{Z}_k)\in\Delta_{X,k}$ can be identified with the $(k-1)$-simplex $\underline{Z}_*^{\widehat{0}}$ of $\Delta^{>\underline{Z}}$. This yields an

isomorphism

$$\mathcal{P}_k := \bigoplus_{\underline{Z}_* \in \Delta_{X,k}} P_{Z_0}^{(\underline{Z}_*)} \cong \bigoplus_{Z \in \mathcal{J}(P)} P_Z \otimes_R C_{k-1}(\Delta^{>\underline{Z}}), \qquad p^{(\underline{Z}_*)} \mapsto \begin{cases} p \otimes \langle \underline{Z}_*^{\widehat{0}} \rangle, & k > 0; \\ p \otimes 1, & k = 0. \end{cases} \tag{3.1.1}$$

Note that, for $Z, T \in \mathcal{J}(P)$ with $\underline{Z} < \underline{T}$ (equivalently, $d(X,T) = d(X,Z) + d(Z,T)$), we have a right A-module homomorphism

$$\mu_{T,Z} \colon P_Z \longrightarrow P_T, \qquad p \longmapsto b_{TZ} p,$$

given by by left-multiplying $b_{TZ} \in e_T A e_Z$. For $k > 0$, define an A-module homomorphism $d_k \colon \mathcal{P}_k \longrightarrow \mathcal{P}_{k-1}$ by

$$d_k\big(p^{(\underline{Z}_*)}\big) := \big(\mu_{Z_1,Z_0}(p)\big)^{(\underline{Z}_*^{\widehat{0}})} + \sum_{i=1}^{k} (-1)^i p^{(\underline{Z}_*^{\widehat{i}})}. \tag{3.1.2}$$

Proposition 3.1. *The maps d_k satisfy $d_{k-1}d_k = 0$ for every $k \geq 2$.*

Proof. Fix $\underline{Z}_* = (\underline{Z}_0 < \cdots < \underline{Z}_k)$. For $0 \leq i \leq k$, write

$$d_k|_{P_{Z_0}^{(\underline{Z}_*)}} = \sum_{i=0}^{k} (-1)^i \varepsilon_i^{\underline{Z}_*},$$

where the homomorphisms $\varepsilon_i^{\underline{Z}_*}$ are

$$\varepsilon_i^{\underline{Z}_*} \colon P_{Z_0}^{(\underline{Z}_*)} \longrightarrow \mathcal{P}_{k-1}, \quad \varepsilon_i^{\underline{Z}_*}\big(p^{(\underline{Z}_*)}\big) := \begin{cases} \big(\mu_{Z_1,Z_0}(p)\big)^{(\underline{Z}_*^{\widehat{0}})}, & i = 0, \\ p^{(\underline{Z}_*^{\widehat{i}})}, & i > 0. \end{cases} \tag{3.1.3}$$

We claim that

$$\varepsilon_i^{\underline{Z}_*^{\widehat{j}}} \varepsilon_j^{\underline{Z}_*} = \varepsilon_{j-1}^{\underline{Z}_*^{\widehat{i}}} \varepsilon_i^{\underline{Z}_*} \qquad (0 \leq i < j \leq k). \tag{3.1.4}$$

Indeed, if $i > 0$, both sides are the identity map between the same two labelled copies of P_{Z_0}. On the other hand, if $i = 0$, then the case when and $j > 1$ implies that both sides are induced by μ_{Z_1,Z_0}. It remains to consider $i = 0$ and $j = 1$. In this case, the two sides are induced respectively by μ_{Z_2,Z_0} and $\mu_{Z_2,Z_1}\mu_{Z_1,Z_0}$. Since $\underline{Z}_0 < \underline{Z}_1 < \underline{Z}_2$, the corresponding distances are additive, and hence $b_{Z_2Z_1}b_{Z_1Z_0} = b_{Z_2Z_0}$. Therefore $\mu_{Z_2,Z_1}\mu_{Z_1,Z_0} = \mu_{Z_2,Z_0}$, which proves (3.1.4).

For every pair $0 \leq i < j \leq k$, the two ways of deleting the i-th and j-th members of $\underline{Z}_*$ occur in $d_{k-1}d_k$ with respective coefficients $(-1)^j(-1)^i = (-1)^{i+j}$ and $(-1)^i(-1)^{j-1} = -(-1)^{i+j}$. By (3.1.4), the corresponding homomorphisms are equal, so these two terms cancel. All terms in $d_{k-1}d_k$ arise uniquely in this way, and therefore $d_{k-1}d_k = 0$. □

Example 3.2. We continue with the example presented in Example 2.15. For a strict chain $\underline{Z}_*$ in $\mathcal{D}_X$, let $P_{Z_0}^{(\underline{Z}_*)}$ be a labeled copy of P_{Z_0}. In this example,

$$\begin{aligned}
\mathcal{P}_0 &= P_{[1]}^{(\underline{\emptyset})} \oplus P_{\emptyset}^{(\underline{1})} \oplus P_{[12]}^{(\underline{2})} \oplus P_{[13]}^{(\underline{3})} \oplus P_{[3]}^{(\underline{13})} \oplus P_{[123]}^{(\underline{23})}, \\
\mathcal{P}_1 &= P_{[1]}^{(\underline{\emptyset}<\underline{1})} \oplus P_{[1]}^{(\emptyset<\underline{2})} \oplus P_{[1]}^{(\underline{\emptyset}<\underline{3})} \oplus P_{[1]}^{(\underline{\emptyset}<\underline{13})} \oplus P_{[1]}^{(\underline{\emptyset}<\underline{23})} \\
&\quad \oplus P_{\emptyset}^{(\underline{1}<\underline{13})} \oplus P_{[12]}^{(\underline{2}<\underline{23})} \oplus P_{[13]}^{(\underline{3}<\underline{13})} \oplus P_{[13]}^{(\underline{3}<\underline{23})}, \\
\mathcal{P}_2 &= P_{[1]}^{(\underline{\emptyset}<\underline{1}<\underline{13})} \oplus P_{[1]}^{(\underline{\emptyset}<\underline{3}<\underline{13})} \oplus P_{[1]}^{(\underline{\emptyset}<\underline{2}<\underline{23})} \oplus P_{[1]}^{(\underline{\emptyset}<\underline{3}<\underline{23})}.
\end{aligned}$$

Note that $\mathcal{P}_{\geq 3} = 0$. For $k > 0$, the differential is

$$d_k\big(p^{(\underline{Z}_*)}\big) = \big(b_{Z_1Z_0}p\big)^{(\underline{Z}_*^{\widehat{0}})} + \sum_{i=1}^{k} (-1)^i p^{(\underline{Z}_*^{\widehat{i}})}.$$

We display the resolution more graphically in Figure 1, where each component of the maps involved are shown more clearly. The labeling on each arrow represent the map, with the unlabeled ones represeting the identity map. We group (highlight) the terms coming from the same subcomplex $\Delta^{>\underline{Z}}$ by the same color.

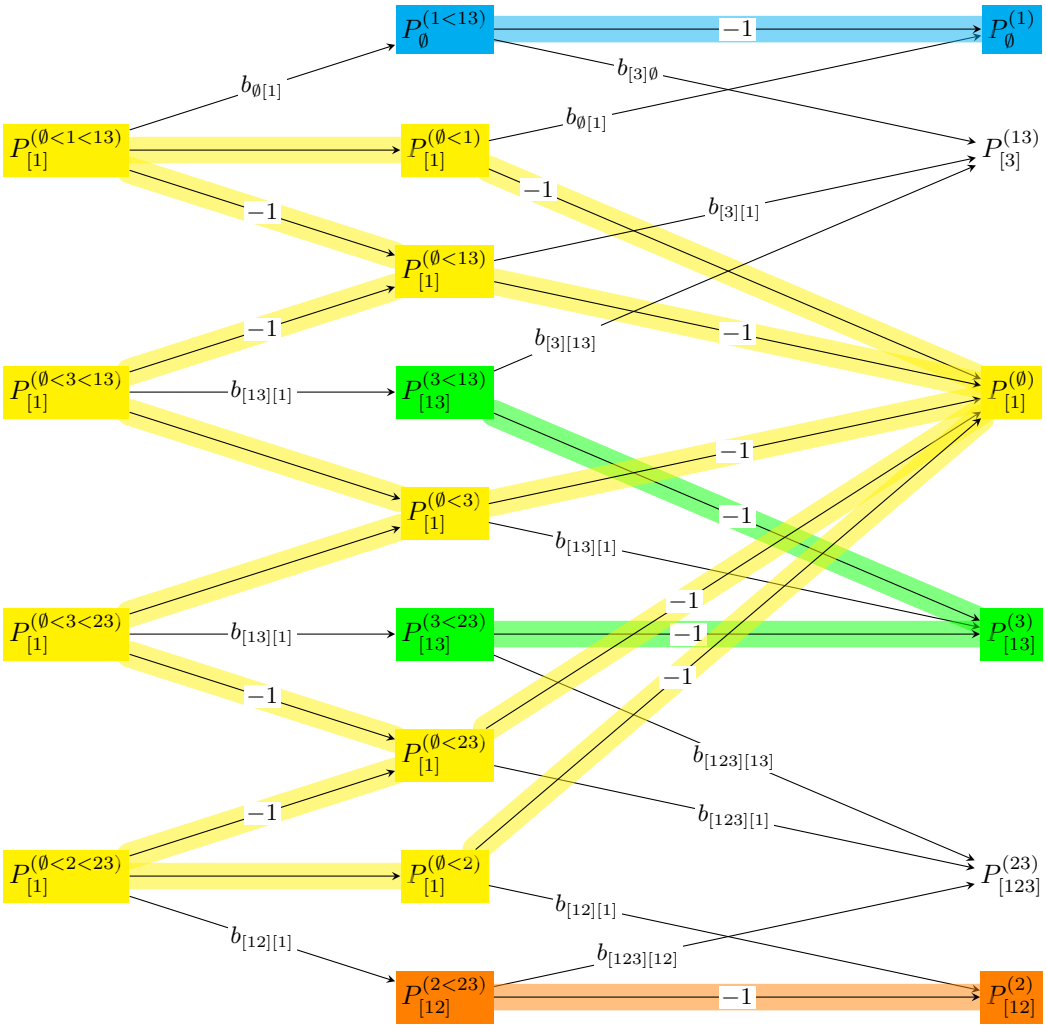


FIGURE 1. $\mathcal{P}_\bullet$ and the contribution from each strict upset subcomplex $\Delta^{>\underline{Z}}$

For each $Y \in \mathcal{J}(P)$, we have a free R-module $I_X e_Y = D(e_Y A e_X)$ of rank 1. Let $\phi_Y \colon P_Y \to I_X$ be the homomorphism determined by sending e_Y to the canonical R-basis element dual to $b_{YX} \in e_Y A e_X$. If $\underline{Z}_0 < \underline{Z}_1$, then the corresponding distances are additive and $b_{Z_1 Z_0} b_{Z_0 X} = b_{Z_1 X}$. Consequently, $\phi_{Z_1} \mu_{Z_1, Z_0} = \phi_{Z_0}$ whenever $\underline{Z}_0 < \underline{Z}_1$. Define

$$d_0 := \big(\phi_Y\big)_{Y \in \mathcal{J}(P)} \colon \mathcal{P}_0 = \bigoplus_{Y \in \mathcal{J}(P)} P_Y \longrightarrow I_X.$$

For $\underline{Z}_0 < \underline{Z}_1$, one has $d_0 d_1 \big(p^{(\underline{Z}_0 < \underline{Z}_1)}\big) = \phi_{Z_1}(b_{Z_1 Z_0} p) - \phi_{Z_0}(p) = 0$. Hence, by Proposition 3.1, we obtain a chain complex

$$\widetilde{\mathcal{P}}_\bullet : \quad \cdots \longrightarrow \mathcal{P}_2 \xrightarrow{d_2} \mathcal{P}_1 \xrightarrow{d_1} \mathcal{P}_0 \xrightarrow{d_0} I_X \longrightarrow 0. \tag{3.1.5}$$

We call $\widetilde{\mathcal{P}}_\bullet$ the *augmented candidate complex* and $\mathcal{P}_\bullet$ is its resolution part.

The main theorem we want to show is the following.

Theorem 3.3. *The complex $\widetilde{\mathcal{P}}_\bullet$ is exact, i.e. $\mathcal{P}_\bullet$ defines a projective resolution of I_X.*

Note that the projective resolution constructed above gives an upper bound on $\operatorname{pdim}_A I_X$ through the lengths of strict chains in $\mathcal{D}_X$. Together with Corollary 2.6, this yields the Iwanaga–Gorenstein property of A.

Counting the lengths of strict chains in $\mathcal{D}_X$ gives the following consequence of Theorem 3.3.

Corollary 3.4. *Assume that $P \neq \emptyset$. Then, for every $X \in \mathcal{J}(P)$,*

$$\operatorname{pdim}_{\Sigma_{\mathcal{J}(P)}} I_X \leq |P| - 1.$$

In particular, if R is Gorenstein, then $\Sigma_{\mathcal{J}(P)}$ is a Gorenstein R-algebra. If R is Iwanaga–Gorenstein, then $\Sigma_{\mathcal{J}(P)}$ is Iwanaga–Gorenstein and

$$\operatorname{idim}(\Sigma_{\mathcal{J}(P)})_{\Sigma_{\mathcal{J}(P)}} = \operatorname{idim}{}_{\Sigma_{\mathcal{J}(P)}}\Sigma_{\mathcal{J}(P)} \leq |P| - 1 + \operatorname{idim}_R R.$$

Proof. Fix $X \in \mathcal{J}(P)$. If $P \setminus X \in \mathcal{J}(P)$, then $I_X \cong P_{P\setminus X}$ by Lemma 2.13, so I_X is projective. Otherwise $P \notin \mathcal{D}_X$, so every strict chain in $\mathcal{D}_X$ has at most $|P|$ members and $\mathcal{P}_k = 0$ for $k \geq |P|$. Theorem 3.3 therefore gives $\operatorname{pdim}_{\Sigma_{\mathcal{J}(P)}} I_X \leq |P| - 1$. The remaining assertions follow from Corollary 2.6. □

Remark 3.5. If $L \cong \mathcal{J}(P)$, then P is isomorphic to the poset of join-irreducible elements of L. Thus the bounds can also be stated in terms of the number of join-irreducible elements of L.

3.2. **Exactness of the candidate complex.** We prove exactness after right multiplication by each vertex idempotent e_W. The first lemma records the multiplication criterion used in both steps of the argument.

Lemma 3.6. *Let $U, V, W \in \mathcal{J}(P)$ with $\underline{U} \leq \underline{V}$. Then $b_{VU}b_{UW} \neq 0 \Leftrightarrow (\underline{V} \setminus \underline{U}) \cap \underline{W} = \emptyset$.*

Proof. We have

$$b_{VU}b_{UW} \neq 0 \iff W \triangle U \subseteq W \triangle V \iff \underline{W} \triangle \underline{Z} \subseteq \underline{W} \triangle \underline{Y},$$

where the first equivalence is Lemma 2.12, the second follows from Lemma 2.11(2). The claim then follows from Lemma 2.11(4) by taking $(X, Y, Z) = (W, V, U)$. □

Fix $W \in \mathcal{J}(P)$. Right multiplication by e_W gives the complex

$$\widetilde{\mathcal{P}}_\bullet e_W : \quad \cdots \longrightarrow \mathcal{P}_1 e_W \longrightarrow \mathcal{P}_0 e_W \longrightarrow I_X e_W \longrightarrow 0.$$

We first replace it by a smaller homotopy equivalent subcomplex.

Proposition 3.7. *Claude points out that this is Lemma 2.9 'with coefficient', which could shrink proof by a half if use the lemma. For $k \geq 0$, put*

$$\mathcal{P}_k^{\leq W} := \bigoplus_{\substack{\underline{Z}_* = (\underline{Z}_0 < \cdots < \underline{Z}_k) \in \Delta_{X,k} \\ \underline{Z}_k \leq \underline{W}}} (P_{Z_0} e_W)^{(\underline{Z}_*)}.$$

Together with $I_X e_W$ in degree -1, these R-modules form a subcomplex $\widetilde{\mathcal{P}}_\bullet^{\leq W}$ of $\widetilde{\mathcal{P}}_\bullet e_W$. Its inclusion

$$\iota_W \colon \widetilde{\mathcal{P}}_\bullet^{\leq W} \longrightarrow \widetilde{\mathcal{P}}_\bullet e_W$$

is a homotopy equivalence.

Proof. Since $P_{Z_0} e_W = R b_{Z_0 W}$, the differential in (3.1.2) evaluates to

$$d_k\big((b_{Z_0W})^{(\underline{Z}_*)}\big) = \big(b_{Z_1 Z_0} b_{Z_0 W}\big)^{(\underline{Z}_*^{\hat{0}})} + \sum_{i=1}^{k} (-1)^i (b_{Z_0 W})^{(\underline{Z}_*^{\hat{i}})}. \tag{3.2.1}$$

Deleting a member of a chain whose largest member is at most $\underline{W}$ preserves this property, so the displayed R-modules form a subcomplex.

For $\underline{Z} \in \mathcal{D}_X$, set $r_W(\underline{Z}) := \underline{Z} \cap \underline{W}$. This again lies in $\mathcal{D}_X$: indeed, recall from Lemma 2.11(3) that $\nabla_X^{Z,W} := (X \cap Z) \cup (X \cap W) \cup (Z \cap W)$, which is also an order ideal, satisfies $\underline{Z} \cap \underline{W} = \underline{\nabla_X^{Z,W}}$. Thus r_W is an order-preserving map from $\mathcal{D}_X$ to $\mathcal{D}_X^{\leq W} := \{\underline{Z} \in \mathcal{J}(P) \mid \underline{Z} \leq \underline{W}\}$.

Fix $\underline{Z}_* = (\underline{Z}_0 < \cdots < \underline{Z}_k)$ and let $T_i := \nabla_X^{Z_i, W}$, so that $\underline{T_i} = \underline{Z_i} \cap \underline{W}$. Define $\rho_{W,-1} := \mathrm{id}_{I_X e_W}$ and, for $k \geq 0$, define

$$\rho_{W,k}\big((b_{Z_0W})^{(\underline{Z}_*)}\big) := \begin{cases} (b_{T_0 W})^{(\underline{T_0} < \cdots < \underline{T_k})}, & \underline{T_0} < \cdots < \underline{T_k}, \\ 0, & \underline{T_i} = \underline{T_{i+1}} \text{ for some } i. \end{cases}$$

Then $\rho_W \iota_W = \mathrm{id}$ as graded maps.

It remains to construct the homotopy. Put $h_{W,-1} := 0$ and, for $k \geq 0$, define

$$h_{W,k}\big((b_{Z_0W})^{(\underline{Z}_*)}\big) := \sum_{j=0}^{k}(-1)^j (b_{T_0W})^{(\underline{T_0}\leq\cdots\leq\underline{T_j}\leq\underline{Z}_j<\cdots<\underline{Z}_k)}.$$

A summand is understood to be zero if two adjacent members of its indexing sequence are equal. We now suppress the subscript W from ι, ρ, and h.

For $k = 0$, a direct calculation gives $d_1h_0 = \mathrm{id}_{\mathcal{P}_0e_W} - \iota_0\rho_0$. For $k \geq 1$, put

$$\tau_j := (\underline{T_0} \leq \cdots \leq \underline{T_j} \leq \underline{Z}_j < \cdots < \underline{Z}_k).$$

Expanding the differential gives

$$\begin{aligned} d_{k+1}h_k\big((b_{Z_0W})^{(\underline{Z}_*)}\big) &= (b_{Z_0T_0}b_{T_0W})^{(\underline{Z}_*)} \\ &\quad + \sum_{j=1}^{k}(-1)^j (b_{T_1T_0}b_{T_0W})^{(\tau_j^{\hat{0}})} \\ &\quad + \sum_{j=0}^{k}\sum_{i=1}^{k+1}(-1)^{i+j}(b_{T_0W})^{(\tau_j^{\hat{i}})}. \end{aligned}$$

Because $(\underline{U} \setminus r_W(\underline{U})) \cap \underline{W} = \emptyset$, Lemma 3.6 gives $b_{Z_iT_i}b_{T_iW} = b_{Z_iW}$. The first term is therefore the original basis element, the single sum is $-h_{k-1}\big((b_{Z_1Z_0}b_{Z_0W})^{(\underline{Z}_*^{\hat{0}})}\big)$, and the double sum is

$$-\sum_{\ell=1}^{k}(-1)^\ell h_{k-1}\big((b_{Z_0W})^{(\underline{Z}_*^{\hat{\ell}})}\big) - \iota_k\rho_k\big((b_{Z_0W})^{(\underline{Z}_*)}\big).$$

Comparing this with the expansion of $h_{k-1}d_k$ yields

$$d_{k+1}h_k + h_{k-1}d_k = \mathrm{id}_{\mathcal{P}_ke_W} - \iota_k\rho_k \qquad (k \geq 0).$$

For $k \geq 1$, applying d_k to this identity and comparing with the identity in degree $k-1$ gives

$$d_k(\mathrm{id} - \iota_k\rho_k) = (\mathrm{id} - \iota_{k-1}\rho_{k-1})d_k.$$

Since ι_W is an injective chain map, it follows that $d_k\rho_k = \rho_{k-1}d_k$. The same conclusion in degree zero follows by applying d_0 to the degree-zero homotopy identity. Thus ρ_W is a chain map and the displayed identity is the required homotopy. □

The lower subcomplex admits a direct decomposition into augmented chain complexes of intervals.

Proposition 3.8. *For every $W \in \mathcal{J}(P)$, the complex $\widetilde{\mathcal{P}}_\bullet^{\leq W}$ is exact.*

Proof. For $\underline{Y} \leq \underline{W}$ and $k \geq 0$, put

$$K_{Y,k} := \bigoplus_{\substack{\underline{Z}_* = (\underline{Y} < \underline{Z}_1 < \cdots < \underline{Z}_k) \in \Delta_{X,k} \\ \underline{Z}_k \leq \underline{W}}} (P_Ye_W)^{(\underline{Z}_*)}.$$

If $\underline{Y} < \underline{Z}_1 \leq \underline{W}$, then $(\underline{Z}_1 \setminus \underline{Y}) \cap \underline{W} \neq \emptyset$. Hence Lemma 3.6 gives $b_{Z_1Y}b_{YW} = 0$, and (3.2.1) shows that $K_{Y,\bullet}$ is a subcomplex.

If $\underline{Y} < \underline{W}$, then the restriction of d_0 to $K_{Y,0}$ is zero by Lemma 2.12. Now let $(\underline{Y}, \underline{W}] := \{\underline{T} \in \mathcal{D}_X \mid \underline{Y} < \underline{T} \leq \underline{W}\}$. Deleting the initial member of an indexing chain in each $K_{Y,k}$ for $k > 0$ and sending $b_{YW}^{(\underline{Y})} \in K_{Y,0}$ to $1 \in C_{-1}(\Delta((\underline{Y}, \underline{W}])))$ defines an isomorphism of chain complexes

$$K_{Y,\bullet} \xrightarrow{\sim} \widetilde{C}_\bullet\big(\Delta((\underline{Y}, \underline{W}])\big)[1].$$

Since $\underline{W}$ is the maximum of $(\underline{Y}, \underline{W}]$, its order complex is a cone and its augmented chain complex is exact.

When $\underline{Y} = \underline{W}$, one has $Y = W$. The complex $K_{W,\bullet}$ is concentrated in degree zero, and $d_0 = \phi_We_W \colon K_{W,0} \longrightarrow I_Xe_W$ is a homomorphism between free R-modules of rank one sending the canonical basis element to the canonical dual basis element, hence an isomorphism. Therefore

$$\widetilde{\mathcal{P}}_\bullet^{\leq W} = \left(\bigoplus_{\underline{Y} < \underline{W}} K_{Y,\bullet}\right) \oplus \left(K_{W,0} \xrightarrow{\sim} I_Xe_W\right)$$

is a direct sum of exact complexes. □

Proof of Theorem 3.3. For every $W \in \mathcal{J}(P)$, Proposition 3.7 identifies $\widetilde{\mathcal{P}}_\bullet e_W$, up to homotopy, with the exact complex $\widetilde{\mathcal{P}}_\bullet^{\leq W}$ from Proposition 3.8. Hence every $\widetilde{\mathcal{P}}_\bullet e_W$ is exact, and so taking the direct sum over $W \in \mathcal{J}(P)$ yields an exact complex $\widetilde{\mathcal{P}}_\bullet = \bigoplus_{W \in \mathcal{J}(P)} \widetilde{\mathcal{P}}_\bullet e_W$. □

Example 3.9. We continue with Examples 2.15 and 3.2. Put $W = [13]$, so $\underline{W} = \{3\}$. Right multiplication by e_W gives

$$\mathcal{P}_\bullet e_W : 0 \longrightarrow \mathcal{P}_2 e_W \xrightarrow{d_2 e_W} \mathcal{P}_1 e_W \xrightarrow{d_1 e_W} \mathcal{P}_0 e_W \xrightarrow{d_0 e_W} I_{[1]} e_W \longrightarrow 0.$$

Explicitly, $\mathcal{P}_k e_W$ is obtained from the displayed expression for $\mathcal{P}_k$ by replacing every labelled copy $P_Z^{(\underline{Z}_*)}$ by $P_Z e_W^{(\underline{Z}_*)}$. The lower subcomplex consists of the labelled copies for which every member of $\underline{Z}_*$ is contained in $\underline{W} = \{3\}$. It is

$$0 \longrightarrow (P_{[1]} e_{[13]})^{(\emptyset < \{3\})} \xrightarrow{d_1} (P_{[1]} e_{[13]})^{(\emptyset)} \oplus (P_{[13]} e_{[13]})^{(\{3\})} \xrightarrow{d_0} I_{[1]} e_{[13]} \longrightarrow 0.$$

For $p \in P_{[1]} e_{[13]}$, its first differential is

$$d_1\big(p^{(\emptyset < \{3\})}\big) = \big(b_{[13],[1]} p\big)^{(\{3\})} - p^{(\emptyset)}.$$

In particular, for the path class $b_{[1],[13]} \in P_{[1]} e_{[13]}$,

$$d_1\big((b_{[1],[13]})^{(\emptyset < \{3\})}\big) = -(b_{[1],[13]})^{(\emptyset)},$$

because $b_{[13],[1]} b_{[1],[13]} = 0$. In the indicated bases, the augmented lower subcomplex is

$$0 \longrightarrow R \xrightarrow{\binom{-1}{0}} R^2 \xrightarrow{(0\ 1)} R \longrightarrow 0.$$

3.3. **Refining projective dimension.** The cases $|P| \leq 2$ are described in Example 2.14. For $|P| \geq 3$, we will improve the bound in Corollary 3.4 by splitting the last differential of $\mathcal{P}_\bullet$. We first record the highest possible projective term, then use the topology of the boundary of a simplex to split its differential.

Lemma 3.10. *Let $X \in \mathcal{J}(P)$ and suppose that $P \setminus X \notin \mathcal{J}(P)$. Then*

$$\Delta_r^{>\underline{X}} = \emptyset \quad \text{for } r \geq |P| - 1, \qquad \mathcal{P}_k = 0 \quad \text{for } k \geq |P|.$$

Moreover, the identification in (3.1.1) restricts to an isomorphism

$$\mathcal{P}_{|P|-1} \cong P_X \otimes_R C_{|P|-2}(\Delta^{>\underline{X}}).$$

Proof. Since $P \setminus X \notin \mathcal{J}(P)$, we have $P \notin \mathcal{D}_X$. Every strict chain of non-empty members of $\mathcal{D}_X$ therefore has at most $|P| - 1$ members, i.e. $\Delta_{|P|-1}^{>\underline{X}} = \emptyset$. This also implies that every strict chain in $\mathcal{D}_X$ has at most $|P|$ members, so $\mathcal{P}_k = 0$ for $k \geq |P|$. For the final isomorphism, note that every chain in $\mathcal{D}_X$ with $|P|$ members begins at $\emptyset = \underline{X}$, so the claim follows from (3.1.1). □

Example 3.11. Continue with $P = \{1 < 2\} \sqcup \{3\}$ and $X = \{1\}$ from Example 2.15. The contribution coming from the complex $\Delta^{>\underline{X}}$ is the yellow-highlighted part in Figure 1. We have $C_1(\Delta^{>\underline{X}}) \cong R^{\oplus 4}$, and so $\mathcal{P}_2 \cong P_X^{\oplus 4}$ and $\mathcal{P}_k = 0$ for $k \geq 3$.

Proposition 3.12. *Assume that $|P| \geq 3$. Then, for every $X \in \mathcal{J}(P)$,*

$$\operatorname{pdim}_{\Sigma_{\mathcal{J}(P)}} I_X \leq |P| - 2.$$

If R is Iwanaga–Gorenstein, then $\Sigma_{\mathcal{J}(P)}$ is Iwanaga–Gorenstein and

$$\operatorname{idim}(\Sigma_{\mathcal{J}(P)})_{\Sigma_{\mathcal{J}(P)}} = \operatorname{idim}{}_{\Sigma_{\mathcal{J}(P)}} \Sigma_{\mathcal{J}(P)} \leq |P| - 2 + \operatorname{idim}_R R.$$

Proof. Fix $X \in \mathcal{J}(P)$. If $P \setminus X \in \mathcal{J}(P)$, then I_X is projective by Lemma 2.13, and the projective dimension bound follows. Assume that $P \setminus X \notin \mathcal{J}(P)$.

We first claim that there is an R-linear map

$$\rho : C_{|P|-3}(\Delta^{>\underline{X}}) \longrightarrow C_{|P|-2}(\Delta^{>\underline{X}})$$

such that $\rho \partial_{|P|-2} = \mathrm{id}$. By Lemma 2.10, it suffices to show that $\Delta^{>\underline{X}}$ is a proper subcomplex of $\Delta(2^P \setminus \{\emptyset, P\})$. Indeed, since $P \setminus X \notin \mathcal{J}(P)$, we have $P \notin \mathcal{D}_X$. By the assumption $P \setminus X \notin \mathcal{J}(P)$ and that $\underline{X} = \emptyset$, every vertex of $\Delta^{>\underline{X}}$ is a non-empty proper subset of P. Another consequence of

$P \setminus X \notin \mathcal{J}(P)$ is that there exist $a < b$ in P with $a \in X$ and $b \in P \setminus X$. Hence for any $T \in \mathcal{J}(P)$, the set $\underline{T} = X \triangle T$ cannot contain both a and b. In particular, $\{a, b\} \notin \mathcal{D}_X$. Since $|P| \geq 3$, the set $\{a, b\}$ is a vertex of $\Delta(2^P \setminus \{\emptyset, P\})$. Thus $\Delta^{>\underline{X}}$ is a proper subcomplex of $\Delta(2^P \setminus \{\emptyset, P\})$, as required.

Now, we will construct the left inverse of $d_{|P|-1} : \mathcal{P}_{|P|-1} \to \mathcal{P}_{|P|-2}$. Consider the projection of $\mathcal{P}_{|P|-2}$ onto the component $P_X \otimes_R C_{|P|-3}(\Delta^{>\underline{Z}})$ via (3.1.1), that is,

$$\pi : \mathcal{P}_{|P|-2} \longrightarrow P_X \otimes_R C_{|P|-3}(\Delta^{>\underline{X}}), \quad p^{(\underline{Z}_*)} \mapsto \begin{cases} p \otimes \langle \underline{Z}_*^{\widehat{0}} \rangle, & Z_0 = X \text{ and } p \in P_{Z_0}, \\ 0, & \text{otherwise.} \end{cases}$$

Consider a chain $\underline{Z}_* = (\underline{Z}_0 < \underline{Z}_1 < \cdots < \underline{Z}_{|P|-1}) \in \mathcal{D}_X$ with $Z_0 = X$ and let $p \in P_X$. Then by definition of the differential (3.1.2) we have

$$\begin{aligned} \pi d_{|P|-1}\big(p^{(\underline{Z}_*)}\big) &= \sum_{i=1}^{|P|-1} (-1)^i p \otimes \langle \underline{Z}_1 < \cdots < \widehat{\underline{Z}_i} < \cdots < \underline{Z}_{|P|-1} \rangle \\ &= -p \otimes \partial_{|P|-2} \langle \underline{Z}_1 < \cdots < \underline{Z}_{|P|-1} \rangle. \end{aligned} \tag{3.3.1}$$

Combining with the isomorphism in Lemma 3.10 we obtain the following composition

$$\mathcal{P}_{|P|-1} \xrightarrow{d_{|P|-1}} \mathcal{P}_{|P|-2} \xrightarrow{\pi} P_X \otimes_R C_{|P|-3}(\Delta^{>\underline{X}}) \xrightarrow{-\operatorname{id}_{P_X} \otimes \rho} P_X \otimes_R C_{|P|-2}(\Delta^{>\underline{X}}) \cong \mathcal{P}_{|P|-1}.$$

Since ρ is the left inverse of $\partial_{|P|-2}$, we get from (3.3.1) that

$$(-\operatorname{id}_{P_X} \otimes \rho)\pi d_{|P|-1} = (-\operatorname{id}_{P_X} \otimes \rho)(-\operatorname{id}_{P_X} \otimes \partial_{|P|-2}) = \operatorname{id}_{P_X} \otimes \operatorname{id}_{C_{|P|-2}(\Delta^{>\underline{X}})},$$

which means that $d_{|P|-1}$ is a split monomorphism. Thus we can truncate the resolution by one more term and get that $\operatorname{pdim}_{\Sigma_{\mathcal{J}(P)}} I_X \leq |P| - 2$. The Iwanaga–Gorenstein assertion and the self-injective dimension bound now follow from Corollary 2.6. □

Remark 3.13. The bound in Proposition 3.12 is optimal: Example 4.7 presents a case when the bound is attained.

4. Morse reduction and the minimal projective resolution

While we now know that $\mathcal{P}_\bullet$ is a projective resolution of I_X, it is far from being minimal. To obtain a minimal resolution, we apply algebraic Morse theory [CLZ26] to remove its invertible differential components. Throughout, we require the ground ring $R = \Bbbk$ to be a field.

4.1. **Algebraic Morse theory.** Roughly speaking, for a complex $(X_\bullet, d)$ that comes with a special gadget $\mathcal{M}$ called Morse matching, algebraic Morse theory says that we can obtain a 'smaller' complex $(X_\bullet^{\mathcal{M}}, d^{\mathcal{M}})$ that is homotopic to (X, d) by identifying terms appearing in a Morse matching appropriately. This usage of algebraic Morse theory we use follows [CLZ26, Section 3].

Definition 4.1. Let $(X_\bullet, d)$ be a complex of right A-modules with fixed decompositions $X_n = \bigoplus_{\alpha \in I_n} X_{n,\alpha}$. Write $d_{\beta,\alpha} \colon X_{n,\alpha} \to X_{n-1,\beta}$ for the matrix components of d_n.

(1) The *weighted quiver* $Q(X_\bullet)$ has the summands $X_{n,\alpha}$ as vertices and an arrow $X_{n,\alpha} \xrightarrow{d_{\beta,\alpha}} X_{n-1,\beta}$ weighted by $d_{\beta,\alpha}$ whenever $d_{\beta,\alpha} \neq 0$.
(2) A *partial matching* $\mathcal{M}$ is a collection of arrows such that
   (a) no two selected arrows share a vertex;
   (b) the weight of every selected arrow is an isomorphism.
(3) Form a new weighted quiver $Q^{\mathcal{M}}$ by replacing each matched arrow $W \xrightarrow{w} U$ by a *dotted* arrow $U \to W$ in reverse direction of weight $-w^{-1}$. A vertex not incident with a matched arrow is called *critical*.
(4) A directed path in $Q^{\mathcal{M}}$ is a *zigzag path* if thick and dotted arrows alternate; a path consisting of one arrow is also allowed. We shall use the following single path notation. For a vertex U in degree n and a vertex V in degree $n-1$, let $\Gamma_n^{\mathcal{M}}(U, V)$ be the set of zigzag paths from U to V of the form

$$U = U_0 \longrightarrow U_1 \dashrightarrow U_2 \longrightarrow \cdots \dashrightarrow U_{2r} \longrightarrow U_{2r+1} = V, \qquad r \geq 0.$$

The weight $\operatorname{wt}(\gamma)$ of a zigzag path γ is the composition, along the order of the zigzag path γ, of the weights of its arrows.

(5) The partial matching $\mathcal{M}$ is a *Morse matching* if it satisfies the following local finiteness hypothesis: for every matched arrow $W \xrightarrow{w} U$, every vertex V in the same degree, say, $n-1$ as U, and every $x \in U$, the sum

$$\sum_{\gamma \in \Gamma_n^{\mathcal{M}}(W,V)} \mathrm{wt}(\gamma)\big(-w^{-1}(x)\big)$$

is well-defined, and it is non-zero for only finitely many vertices V.

One special case where the the local finiteness hypothesis can be conveniently verified is the following.

Proposition 4.2. ([CLZ26, Proposition 3.2]) *Let $\mathcal{M}$ be a partial matching of $Q(X_\bullet)$. If $Q^{\mathcal{M}}$ contains no infinite zigzag path, then $\mathcal{M}$ is a Morse matching.*

We next record the main result of algebraic Morse theory.

Theorem 4.3. ([CLZ26, Theorem 3.3]) *Let $\mathcal{M}$ be a Morse matching of $Q(X_\bullet)$, and write $\mathrm{Crit}_n(\mathcal{M})$ for its critical vertices in degree n. Then we have complex $(X_\bullet^{\mathcal{M}}, d^{\mathcal{M}})$, called Morse complex, given by*

$$X_n^{\mathcal{M}} := \bigoplus_{U \in \mathrm{Crit}_n(\mathcal{M})} U, \qquad d_{V,U}^{\mathcal{M}} := \sum_{\gamma \in \Gamma_n^{\mathcal{M}}(U,V)} \mathrm{wt}(\gamma),$$

where $d_{V,U}^{\mathcal{M}}$ is the (V,U)-component of $d_n^{\mathcal{M}} : X_n^{\mathcal{M}} \to X_{n-1}^{\mathcal{M}}$ for $U \in \mathrm{Crit}_n(\mathcal{M})$ and $V \in \mathrm{Crit}_{n-1}(\mathcal{M})$. Moreover, it is homotopy equivalent to $(X_\bullet, d)$ – more precisely, one has a decomposition of chain complexes

$$(X_\bullet, d) \cong (X_\bullet^{\mathcal{M}}, d^{\mathcal{M}}) \oplus (Y_\bullet, d^Y),$$

where $Y_\bullet$ is null-homotopic.

In particular, if $X_\bullet$ is a projective resolution of a module, then its Morse complex is also a projective resolution of the same module.

4.2. **Construction of the Morse matching.** Let us first explain the idea of how to find a Morse matching. The differential formula (3.1.2) separates the faces of a strict chain into two types: the zeroth face deletes its initial member, whereas the remaining faces preserve the initial member. Under the identification (3.1.1), these two contributions give a decomposition $d = \delta + \partial$, where the non-zero components of ∂ are signed identity maps between copies of the same projective, while those of δ is induced by the left-multiplication map (more precise formulation will come after the next paragraph).

The difficulty of finding a Morse matching is that the signed identity components of ∂ need not themselves form a partial matching, since several of them may share a source or target (see Example 3.2). The 'solution' is to do what usual homological perturbation theory does: decompose using a choice of kernel–image splitting for each chain complex $C_\bullet(\Delta^{>\underline{Z}})$. With respect to this new choice of decomposition, the matrix components of both ∂ and δ will change, but ∂ becomes a collection of isomorphisms with pairwise disjoint endpoints, whereas every matrix component arising from δ are still radical maps. The former components will define the Morse matching, and the latter property will control its zigzag paths.

We now construct the matching. All tensor products below are over $\Bbbk$. For $Z \in \mathcal{J}(P)$, put

$$C_{Z,\bullet} := \widetilde{C}_\bullet(\Delta^{>\underline{Z}})[1], \qquad C_{Z,k} = \widetilde{C}_{k-1}(\Delta^{>\underline{Z}}), \qquad \partial_k^Z : C_{Z,k} \longrightarrow C_{Z,k-1},$$

where $\partial_k^Z = -\partial_{k-1}$ is the negative of the boundary map on $\widetilde{C}_\bullet(\Delta^{>\underline{Z}})$. Thus $C_{Z,0} = \Bbbk 1_Z$, $C_{Z,k} = 0$ for $k < 0$, and $\partial_0^Z = 0$.

For $\underline{Z} < \underline{T}$ and $k \geq 1$, define $\delta_{T,Z,k} : C_{Z,k} \to C_{T,k-1}$ on the standard basis by

$$\delta_{T,Z,k}\langle \underline{Z}_1 < \cdots < \underline{Z}_k \rangle = \begin{cases} 1_T, & T = Z_1 \text{ and } k = 1, \\ \langle \underline{Z}_2 < \cdots < \underline{Z}_k \rangle, & T = Z_1 \text{ and } k > 1, \\ 0, & T \neq Z_1. \end{cases}$$

Note that here we have $\underline{Z} < \underline{Z}_1 < \cdots < \underline{Z}_k$; when $T = Z_1$, the output is a basis element of $C_{T,k-1}$. Under (3.1.1), we have, for $k \geq 1$,

$$d_k = \partial_k + \delta_k : \mathcal{P}_k \longrightarrow \mathcal{P}_{k-1},$$
$$\partial_k := \bigoplus_Z \mathrm{id}_{P_Z} \otimes \partial_k^Z, \qquad \delta_k := \sum_{\underline{Z}<\underline{T}} \mu_{T,Z} \otimes \delta_{T,Z,k}.$$

The (T, Z)-component of δ_k has source $P_Z \otimes C_{Z,k}$ and target $P_T \otimes C_{T,k-1}$. Recall that $\mu_{T,Z} : P_Z \to P_T$ is given by $p \mapsto b_{TZ}p$. Note that .

For $k \geq 1$, let $\underline{Z}_* = (\underline{Z} < \underline{Z}_1 < \cdots < \underline{Z}_k)$ be a chain in $\mathcal{D}_X$ and let $p \in P_Z$. Under the identification (3.1.1), the element $p^{(\underline{Z}_*)}$ corresponds to $p \otimes \langle \underline{Z}_1 < \cdots < \underline{Z}_k \rangle \in P_Z \otimes C_{Z,k}$. not needed here, the formulation at this point does not use $P_Z^{(\underline{Z}_*)}$

Since $\Bbbk$ is a field, choose, for every Z and $k \geq 0$, a decomposition

$$C_{Z,k} = B_{Z,k} \oplus H_{Z,k} \oplus L_{Z,k}, \quad \text{where} \begin{cases} B_{Z,k} := \operatorname{Im} \partial_{k+1}^Z, \\ B_{Z,k} \oplus H_{Z,k} = \operatorname{Ker} \partial_k^Z, \\ \partial_k^Z\big|_{L_{Z,k}} : L_{Z,k} \xrightarrow{\sim} B_{Z,k-1}, \end{cases} \quad \text{for all } k \geq 1 \tag{4.2.1}$$

In particular, $L_{Z,0} = 0$, and the canonical projection from $\operatorname{Ker} \partial_k^Z$ restricts to an isomorphism

$$H_{Z,k} \xrightarrow{\sim} H_k(C_{Z,\bullet}) \cong \widetilde{H}_{k-1}(\Delta^{>\underline{Z}}), \qquad h \longmapsto h + B_{Z,k}.$$

Use the resulting decomposition

$$\mathcal{P}_k = \bigoplus_{Z \in \mathcal{J}(P)} \big((P_Z \otimes B_{Z,k}) \oplus (P_Z \otimes H_{Z,k}) \oplus (P_Z \otimes L_{Z,k}) \big)$$

to form the weighted quiver of the resolution part $\mathcal{P}_\bullet$, that is, the vertices of the weighted quiver at degree k are given by (Z, E) for $Z \in \mathcal{J}(P)$ and $E \in \{B, H, L\}$ such that the summand $P_Z \otimes E_{Z,k}$ is non-zero. Note that the map ∂_k sends $P_Z \otimes C_{Z,k}$ into $P_Z \otimes C_{Z,k-1}$, whereas δ_k sends it into $\bigoplus_{\underline{Z}<\underline{T}} P_T \otimes C_{T,k-1}$.

Theorem 4.4. *Suppose the ground ring is a field $R = \Bbbk$. With respect to the decomposition (4.2.1), define $\mathcal{M}$ to be the collection of all nonzero components of*

$$\mathrm{id}_{P_Z} \otimes \partial_k^Z\big|_{L_{Z,k}} : P_Z \otimes L_{Z,k} \xrightarrow{\sim} P_Z \otimes B_{Z,k-1} \tag{4.2.2}$$

for all $Z \in \mathcal{P}$ and all $k \geq 1$. Then $\mathcal{M}$ is a Morse matching, and its associated Morse complex defines a minimal projective resolution

$$\cdots \longrightarrow \mathcal{P}_{\min,2} \longrightarrow \mathcal{P}_{\min,1} \longrightarrow \mathcal{P}_{\min,0} \longrightarrow I_X \longrightarrow 0$$

whose terms, for $k \geq 0$, are

$$\mathcal{P}_{\min,k} = \bigoplus_{Z \in \mathcal{J}(P)} P_Z \otimes H_{Z,k} \cong \bigoplus_{Z \in \mathcal{J}(P)} P_Z \otimes \widetilde{H}_{k-1}(\Delta^{>\underline{Z}}) \cong \bigoplus_{Z \in \mathcal{J}(P)} P_Z^{\oplus \widetilde{\beta}_{Z,k}}, \tag{4.2.3}$$

where $\widetilde{\beta}_{Z,k} := \dim_{\Bbbk} \widetilde{H}_{k-1}(\Delta^{>\underline{Z}})$.

Proof. To see that $\mathcal{M}$ is a partial matching, notice first that both the source $P_Z \otimes L_{Z,k}$ and the target $P_Z \otimes B_{Z,k-1}$ of each displayed map have label Z. Since every nonzero component of δ_k changes the label from Z to some T with $\underline{Z} < \underline{T}$, the component of δ_k between these two summands is zero. Hence the component of $d_k = \partial_k + \delta_k$ from $P_Z \otimes L_{Z,k}$ to $P_Z \otimes B_{Z,k-1}$ is exactly $\mathrm{id}_{P_Z} \otimes \partial_k^Z\big|_{L_{Z,k}}$. Their endpoints are pairwise disjoint and their weights are isomorphisms, so $\mathcal{M}$ is a partial matching.

To show that $\mathcal{M}$ is a Morse matching, it follows from Proposition 4.2 that showing $\mathcal{M}$ has no infinite zigzag path will suffice. Indeed, since every nonzero component of ∂ belongs to $\mathcal{M}$, every unmatched arrow in $Q^{\mathcal{M}}$ is a component of δ. Thus, for $k \geq 1$, every unmatched arrow has the form

$$P_Z \otimes E_{Z,k} \xrightarrow{\mu_{T,Z} \otimes \delta'_{T,Z}} P_T \otimes F_{T,k-1} \quad \text{with } \underline{Z} < \underline{T},$$

for appropriate $E, F \in \{B, H, L\}$ and the map $\delta'_{T,Z}$ is given by the composition

$$\delta'_{T,Z} : E_{Z,k} \hookrightarrow C_{Z,k} \xrightarrow{\delta_{T,Z,k}} C_{T,k-1} \twoheadrightarrow F_{T,k-1}.$$

On the other hand, every dotted arrow goes from $P_Z \otimes B_{Z,k-1}$ to $P_Z \otimes L_{Z,k}$. Hence, along a zigzag path, dotted arrows preserve the label $\underline{Z} \in \mathcal{D}_X$, whereas unmatched arrows strictly increase it. Since $\mathcal{D}_X$ is finite and the two types of arrows alternate, zigzag paths have bounded length. In particular, there is no infinite zigzag path, as required.

Theorem 4.3 yields a Morse complex $(\mathcal{P}_{\min,\bullet}, d_{\min})$ homotopy equivalent to $\mathcal{P}_\bullet$. Its terms are projective, and $\mathcal{P}_\bullet$ resolves I_X by Theorem 3.3. Hence $\mathcal{P}_{\min,\bullet}$ is a projective resolution of I_X. To see that the individual terms are as described in (4.2.3), recall from our choice of decomposition (4.2.1) and $\mathcal{M}$ that every nonzero summand $P_Z \otimes B_{Z,k}$ is matched with $P_Z \otimes L_{Z,k+1}$. Thus the critical summands in degree k are precisely the nonzero $P_Z \otimes H_{Z,k}$. Together with $H_{Z,k} \cong \widetilde{H}_{k-1}(\Delta^{>\underline{Z}})$ yields the required description (4.2.3).

It remains to prove minimality. Note that every unmatched arrow has weight $\mu_{T,Z} \otimes \delta'_{T,Z} : P_Z \otimes E_{Z,k} \to P_T \otimes F_{T,k-1}$, where $k \geq 1$ and $\underline{Z} < \underline{T}$, with appropriate $E, F \in \{B, H, L\}$. Since $Z \neq T$, we have $b_{TZ} \in e_T \operatorname{rad} \Sigma_{\mathcal{J}(P)}$, which implies that the map $\mu_{T,Z}$ given by left-multiplying b_{TZ} is radical. Hence, we have

$$\operatorname{Im}(\mu_{T,Z} \otimes \delta'_{T,Z}) \subseteq (P_T \operatorname{rad} \Sigma_{\mathcal{J}(P)}) \otimes F_{T,k-1} = \operatorname{rad} P_T \otimes F_{T,k-1}. \tag{4.2.4}$$

Fix $k \geq 1$ and a zigzag path γ from $P_Z \otimes H_{Z,k}$ to $P_T \otimes H_{T,k-1}$ contributing to $d_{\min,k}$. Since the target of γ is critical, the last arrow of γ has to be unmatched. The image of $\operatorname{wt}(\gamma)$ is contained in the image of this last arrow, so the inclusion (4.2.4) applied to $F = H$ gives $\operatorname{Im} \operatorname{wt}(\gamma) \subseteq \operatorname{rad} P_T \otimes H_{T,k-1}$. By Theorem 4.3, each matrix component of $d_{\min,k}$ is a sum of these path weights; the sum is finite because $Q^{\mathcal{M}}$ is finite and its zigzag paths have bounded length. Hence $\operatorname{Im} d_{\min,k} \subseteq \operatorname{rad} \mathcal{P}_{\min,k-1}$ for every $k \geq 1$, which proves minimality. □

The choices in (4.2.1) may change the individual differentials in the Morse complex. Nevertheless, the resulting complexes are minimal projective resolutions of the same module and are therefore isomorphic. In particular, P_Z occurs in degree k precisely when $\widetilde{H}_{k-1}(\Delta^{>\underline{Z}}) \neq 0$.

Remark 4.5. Note that the multiplicity of P_Z appearing in the minimal projective resolution of I_X is given by $\dim_{\Bbbk} \operatorname{Ext}^k_A(I_X, S_Z)$, and so we have

$$\widetilde{\beta}_{Z,k} = \dim_{\Bbbk} \widetilde{H}_{k-1}(\Delta^{>\underline{Z}}) = \dim_{\Bbbk} \operatorname{Ext}^k_A(I_X, S_Z).$$

We can now refine Corollary 3.4 when working over a field.

Corollary 4.6. *Assume that the ground ring is a field $R = \Bbbk$ and $P \neq \emptyset$. For every $X \in \mathcal{J}(P)$, the indecomposable injective module I_X over the distance $\Bbbk$-algebra Σ_L satisfies*

$$\operatorname{pdim}_{\Sigma_L} I_X = \max\left\{k \geq 0 \;\middle|\; \widetilde{H}_{k-1}(\Delta^{>\underline{Z}}) \neq 0 \text{ for some } Z \in \mathcal{J}(P)\right\} \tag{4.2.5}$$

In particular, for every $Z \in \mathcal{J}(P)$, $\widetilde{\beta}_{Z,k} = \dim_{\Bbbk} \widetilde{H}_{k-1}(\Delta^{>\underline{Z}}) = 0$ if any of the following is satisfied

1. $k = |P|$.
2. $k = |P| - 1$ *and* $|P| \geq 3$.

Proof. The projective dimension formula follows from the minimal terms in (4.2.3). The Betti number vansihing follows from combining Corollary 3.4 and Proposition 3.12 with (4.2.5). □

Example 4.7. Continue with Examples 2.15 and 3.2. We can regard Figure 1 as the weighted quiver with respect to the standard decomposition of $\mathcal{P}_k$ into indecmposable direct summands, which are indexed by chains in $\Delta(\mathcal{D}_X)$. Let us demonstrate the alternative choice used in Theorem 4.4.

For $Z = [1]$, take

$$\begin{aligned} L_{[1],2} = C_{[1],2} &= \Bbbk\{\langle \underline{1} < \underline{13}\rangle, \langle \underline{3} < \underline{13}\rangle, \langle \underline{3} < \underline{23}\rangle, \langle \underline{2} < \underline{23}\rangle\}, \\ B_{[1],1} &= \Bbbk\{\langle \underline{1}\rangle - \langle \underline{13}\rangle, \langle \underline{3}\rangle - \langle \underline{13}\rangle, \langle \underline{3}\rangle - \langle \underline{23}\rangle, \langle \underline{2}\rangle - \langle \underline{23}\rangle\}, \\ L_{[1],1} &= \Bbbk\{\langle \underline{13}\rangle\}. \end{aligned}$$

For the remaining sets with non-zero degree one terms, take

$$\begin{aligned} &L_{\emptyset,1} = \Bbbk\{\langle \underline{13}\rangle\}, \qquad L_{[12],1} = \Bbbk\{\langle \underline{23}\rangle\}, \qquad L_{[13],1} = \Bbbk\{\langle \underline{13}\rangle\}, \qquad H_{[13],1} = \Bbbk\{\langle \underline{13}\rangle - \langle \underline{23}\rangle\}, \\ &B_{Z,0} = C_{Z,0} \quad \text{for } Z \in \{[1], \emptyset, [12], [13]\}, \\ &H_{Z,0} = C_{Z,0} \quad \text{for } Z \in \{[3], [123]\}. \end{aligned}$$

All unlisted spaces in the decomposition (4.2.1) are zero.

The restriction of the internal differential to $L_{[1],2}$ is given by

$$\langle \underline{1} < \underline{13}\rangle \longmapsto \langle \underline{1}\rangle - \langle \underline{13}\rangle, \qquad \langle \underline{3} < \underline{13}\rangle \longmapsto \langle \underline{3}\rangle - \langle \underline{13}\rangle,$$
$$\langle \underline{3} < \underline{23}\rangle \longmapsto \langle \underline{3}\rangle - \langle \underline{23}\rangle, \qquad \langle \underline{2} < \underline{23}\rangle \longmapsto \langle \underline{2}\rangle - \langle \underline{23}\rangle.$$

Moreover, ∂_1^Z sends the displayed generator of each $L_{Z,1}$ to $-1 \in B_{Z,0}$. These eight isomorphisms have pairwise disjoint endpoints and are selected for the Morse matching. In terms of the colors in Figure 1, the yellow part gives four matching components from degree two to degree one and one from degree one to degree zero, while the blue, green, and orange parts each give one matching component from degree one to degree zero. In particular, the two green arrows have the same target and both have weight -1. After change of basis, the green part therefore contributes one matching component from $P_{[13]} \otimes L_{[13],1}$ to $P_{[13]} \otimes B_{[13],0}$, while the difference $\langle \underline{13}\rangle - \langle \underline{23}\rangle$ spans $H_{[13],1}$ and determines the critical summand $P_{[13]} \otimes H_{[13],1}$.

The matrix components of δ also change. For example, $\delta_2\big(p \otimes \langle \underline{3} < \underline{23}\rangle\big) = b_{[13][1]}p \otimes \langle \underline{23}\rangle$, and, in the chosen decomposition of $C_{[13],1}$, we get that $\langle \underline{23}\rangle = \langle \underline{13}\rangle - \big(\langle \underline{13}\rangle - \langle \underline{23}\rangle\big)$. It thus produces two matrix components, of weights $b_{[13][1]}$ and $-b_{[13][1]}$, respectively. Similarly,

$$\delta_1\big(p \otimes (\langle \underline{3}\rangle - \langle \underline{13}\rangle)\big) = b_{[13][1]}p \otimes 1 - b_{[3][1]}p \otimes 1.$$

These are precisely the corresponding arrows displayed in Figure 2, where we write $P_Z\langle c\rangle := P_Z \otimes_{\Bbbk} \Bbbk c$ for compactness.

In Figure 2, the thick blue arrows form the Morse matching, the thin gray arrows are the remaining non-zero matrix components of the differential, and the yellow vertices are critical points.

The two summands $P_{[3]}$ and $P_{[123]}$ without any highlight in Figure 1 are precisely the degree-zero terms of the resulting Morse complex. Its degree-one term is the copy of $P_{[13]}$ represented in the original basis by $p^{(\underline{3}<\underline{13})} - p^{(\underline{3}<\underline{23})}$, for $p \in P_{[13]}$. For the displayed choice of splitting, the remaining differential is

$$d_{\min,1}\colon P_{[13]} \longrightarrow P_{[3]} \oplus P_{[123]}, \qquad p \longmapsto \big(b_{[3][13]}p, -b_{[123][13]}p\big).$$

Note that this example shows that the bound $|P| - 2 = 1$ in Proposition 3.12 is attained.

4.3. **Projective dimension and Castelnuovo–Mumford regularity.** The formula for $\operatorname{pdim} I_X$ bears a striking resemblance to the characterisation of Castelnuovo–Mumford regularity by reduced homology:

$$\operatorname{pdim} I_X = \max\{k \geq 0 : \widetilde{H}_{k-1}(\Delta^{>\underline{Z}}) \neq 0 \text{ for some } Z \in L\}, \tag{4.3.1}$$

$$\operatorname{reg} \Bbbk[K] = \max\{k \geq 0 : \widetilde{H}_{k-1}(\operatorname{lk}_K F) \neq 0 \text{ for some } F \in K\}. \tag{4.3.2}$$

Here K is a finite simplicial complex, $\Bbbk[K]$ is its Stanley–Reisner ring and reg denotes the Castelnuovo–Mumford regularity (Definition 4.12), and $\operatorname{lk}_K F$ is the link of F in K (Definition 2.7). For the equality (4.3.2), it is an application (Lemma 4.13) of a more well-known Hochster's formula. This motivates us to seek a complex K for which we can associate to each $\Delta^{>\underline{Z}}$ a homologous link $\operatorname{lk}_K F$.

We explain the idea of how to seek a suitable K. First, the most natural choice of K should be a refinement of $\mathcal{D}_X$ since we hope every $\Delta^{>\underline{Z}}$ can appear as a link (or at least homologous to it), and it seems to be the nicest case is to have $\underline{Z}$ as a face. Next, if we approach from the link side, then it seems natural to replace the link in (4.3.2) by the order complex of its non-empty-face poset $\mathcal{F}^{\times}(\operatorname{lk}_K F)$, so that we get ourselves to align as close as possible to $\Delta^{>\underline{Z}}$. More precisely, Lemma 2.8 gives $\widetilde{H}_k(\Delta(\mathcal{F}^{\times}(\operatorname{lk}_K F))) \cong \widetilde{H}_k(\operatorname{lk}_K F)$ for all k.

Now that we have order complexes of two different posets, we can then use order homotopy thoerem [Bjö95, Theorem 10.11] to see if their order complexes have the same reduced homology by just finding order-preserving maps. Since we are looking at faces (i.e. sets) containing F, and we want $F = \underline{Z}$, and we also want an order-preserving map from $(\underline{Z}, \hat{1}]$ to a face poset, these lead us to the following choice of K:

$$\mathcal{S}_X := \{S \subseteq P \mid S \subseteq \underline{T} \text{ for some } \underline{T} \in \mathcal{D}_X\} = \bigcup_{\underline{T} \in \max \mathcal{D}_X} 2^{\underline{T}},$$

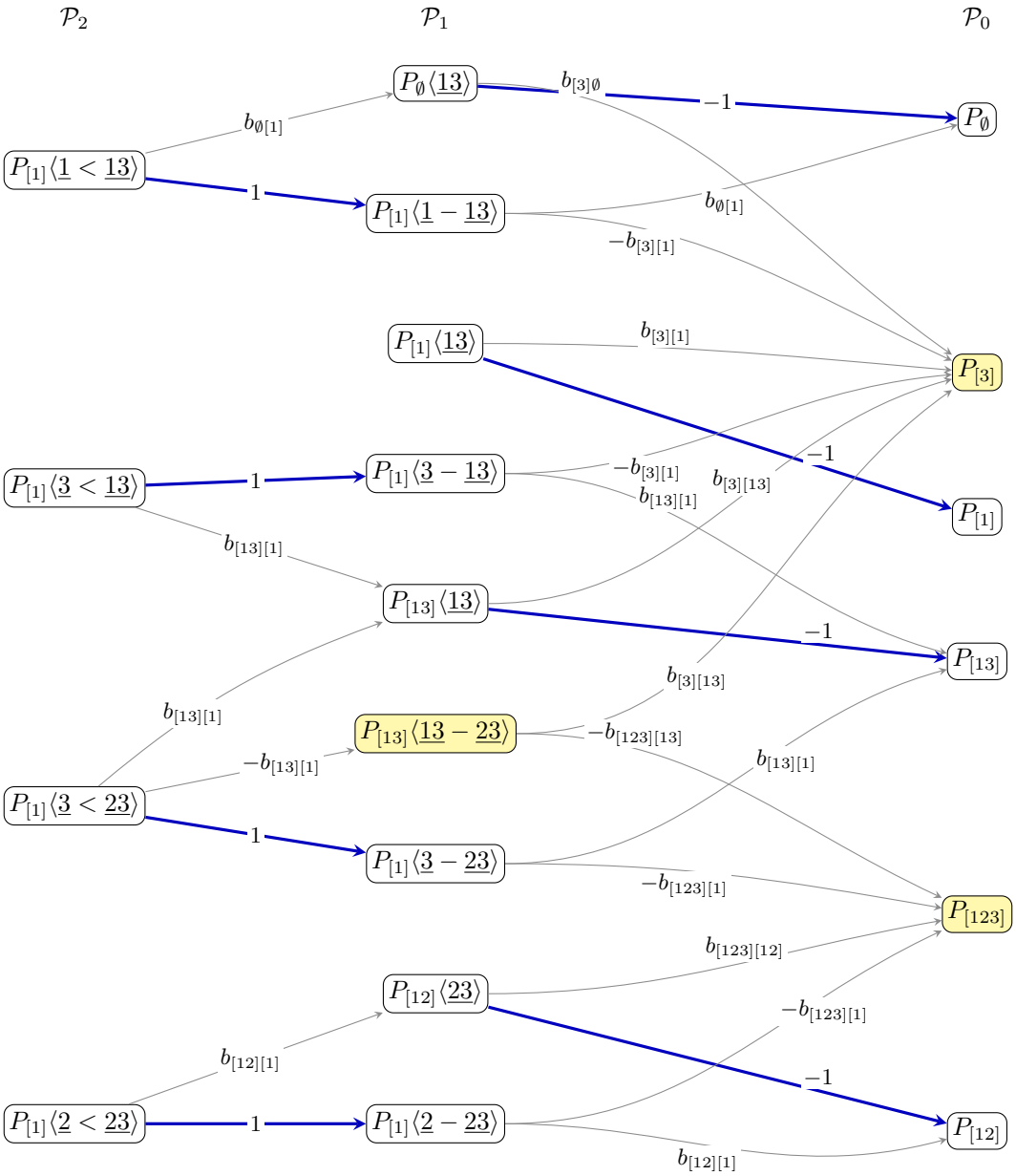


FIGURE 2. The weighted quiver after the $B \oplus H \oplus L$ change of basis.

Indeed, a face in $\mathcal{S}_X$ containing some $\underline{Z}$ tells us possible choices to extend $\underline{Z}$ to a chain, which feels very close to what $\Delta^{>\underline{Z}}$ describes. Our main result of this subsection is the following.

Theorem 4.8. *Let $\Bbbk$ be a field, $\Sigma_{\mathcal{J}(P)}$ be the distance algebra associated to the lattice $\mathcal{J}(P)$ defined over $\Bbbk$. Then, for any $X \in \mathcal{J}(P)$, we have*

$$\operatorname{pdim}_A I_X = \operatorname{reg} \Bbbk[\mathcal{S}_X].$$

In fact, the first step of the proof is to show that $\widetilde{H}_r(\Delta^{>\underline{Z}}) \cong \widetilde{H}_r(\operatorname{lk}_{\mathcal{S}_X} \underline{Z})$ for all $r \geq -1$, which gives us $\operatorname{pdim} I_X \leq \operatorname{reg} \Bbbk[\mathcal{S}_X]$. To get the equality, the second step is to show that $\operatorname{lk}_{\mathcal{S}_X} F$ has zero reduced homology whenever $F \in \mathcal{S}_X \setminus \mathcal{D}_X$. Both steps use the fact that $\mathcal{D}_X$ is closed under intersection.

We use the topological preliminaries from Subsection 2.4 to compare $\mathcal{D}_X$ and $\mathcal{S}_X$.

Lemma 4.9. *The poset $\mathcal{D}_X$ is closed under intersections of non-empty finite families. The map*

$$c : \mathcal{F}(\mathcal{S}_X) \longrightarrow \mathcal{D}_X, \qquad F \longmapsto \bigcap_{\substack{\underline{Y} \in \max \mathcal{D}_X \\ F \subseteq \underline{Y}}} \underline{Y}$$

is well-defined, preserves inclusion, and satisfies $F \subseteq c(F)$.

Proof. For $Y, Z \in \mathcal{J}(P)$, the we have an order ideal $\nabla_X^{Y,Z}$ so that $\underline{\nabla_X^{Y,Z}} = \underline{Y} \cap \underline{Z}$ by Lemma 2.11(3). Hence, closure under intersections of non-empty finite families follows by induction.

Every $F \in \mathcal{S}_X$ is contained in a facet, so the family defining $c(F)$ is non-empty. Closure under intersections gives $c(F) \in \mathcal{D}_X$, and every set in the intersection contains F. If $F \subseteq G$, the family of facets containing G is contained in the family containing F. Therefore $c(F) \subseteq c(G)$. □

Lemma 4.10. *For every $Z \in \mathcal{J}(P)$ and every $k \geq -1$,*

$$\widetilde{H}_k(\Delta^{>\underline{Z}}) \cong \widetilde{H}_k\big(\mathrm{lk}_{\mathcal{S}_X}(\underline{Z})\big).$$

Proof. The set $\underline{Z}$ is a face of $\mathcal{S}_X$. Consider the order-preserving maps

$$j_Z : (\underline{Z}, \hat{1}] \longrightarrow \mathcal{F}(\mathcal{S}_X)_{>\underline{Z}}, \qquad \underline{T} \longmapsto \underline{T},$$

$$c_Z : \mathcal{F}(\mathcal{S}_X)_{>\underline{Z}} \longrightarrow (\underline{Z}, \hat{1}], \qquad F \longmapsto c(F).$$

The map c_Z is well-defined because $\underline{Z} < F \subseteq c(F)$ by Lemma 4.9. Moreover, we have

$$\underline{T} \subset c_Z j_Z(\underline{T}), \text{ for all } \underline{T} \in (\underline{Z}, \hat{1}] \text{ and } F \subseteq j_Z c_Z \text{ for all } F \in \mathcal{F}(\mathcal{S}_X)_{>\underline{Z}}.$$

Hence, Lemma 2.9 says that $j_{Z,\#}$ and $c_{Z,\#}$ are chain homotopy inverses. Thus, the order complexes of $(\underline{Z}, \hat{1}]$ and $\mathcal{F}(\mathcal{S}_X)_{>\underline{Z}}$ have isomorphic reduced homology. Now applies Lemma 2.8 and get that

$$\widetilde{H}_k(\Delta^{>\underline{Z}}) \cong \widetilde{H}_k(\Delta\big(\mathcal{F}(\mathcal{S}_X)_{\underline{Z}}\big)) \cong \widetilde{H}_k(\mathrm{lk}_{\mathcal{S}_X} \underline{Z}) \text{ for all } k \geq -1,$$

which proves the claim. □

Now we need to look at the reduce homologies of links of faces $F \in \mathcal{S}_X \setminus \mathcal{D}_X$.

Lemma 4.11. *If $F \in \mathcal{S}_X$ and $F \notin \mathcal{D}_X$, then $\mathrm{lk}_{\mathcal{S}_X}(F)$ is a cone. In particular,*

$$\widetilde{H}_i\big(\mathrm{lk}_{\mathcal{S}_X}(F)\big) = 0 \qquad \textit{for every } i.$$

Proof. Since $c(F) \in \mathcal{D}_X$ and $F \notin \mathcal{D}_X$, the inclusion $F \subseteq c(F)$ is strict. Choose $p \in c(F) \setminus F$.

For $G \in \mathrm{lk}_{\mathcal{S}_X}(F)$, choose a facet $\underline{Y}$ containing $F \cup G$. The definition of $c(F)$ gives $p \in \underline{Y}$, so $F \cup G \cup \{p\} \subseteq \underline{Y}$. Since $p \notin F$, it follows that $G \cup \{p\} \in \mathrm{lk}_{\mathcal{S}_X}(F)$. Thus the link is a cone with vertex p. □

Let us review the necessary background we need about Stanley–Reisner ring before proving Theorem 4.8.

Definition 4.12. Let K be a finite simplicial complex and put $S_K := \Bbbk[x_v \mid v \in V(K)]$, with $\deg x_v = 1$. Its *Stanley–Reisner ideal* and *Stanley–Reisner ring* are

$$I(K) := \left(\prod_{v \in F} x_v \;\middle|\; F \subseteq V(K),\ F \notin K \right), \qquad \Bbbk[K] := S_K / I(K).$$

Thus a product of distinct variables is non-zero in $\Bbbk[K]$ precisely when its support is a face of K.

Write the i-th term of the minimal graded free resolution of $\Bbbk[K]$ over S_K as $\bigoplus_j S_K(-j)^{\oplus \beta_{i,j}}$, where $(-j)$ denotes the grading shift so that $1 \in S_K(-j)$ is in degree j. The *Castelnuovo–Mumford regularity* is

$$\operatorname{reg} \Bbbk[K] := \max\{j - i \mid \beta_{i,j} \neq 0\}.$$

For $K = \{\emptyset\}$, set $S_K = \Bbbk$, so $\operatorname{reg} \Bbbk[K] = 0$.

Lemma 4.13. *For every finite simplicial complex K,*

$$\operatorname{reg} \Bbbk[K] = \max\{k \geq 0 \mid \widetilde{H}_{k-1}(\mathrm{lk}_K(F)) \neq 0 \textit{ for some } F \in K\}.$$

Proof. This is a reformulation of Hochster's formula and interpretation of Castelnuovo–Mumford regularity using local cohomology; we clarify the reason for completeness and refer unexplained terminologies and references to [HH11, Section A.7]. Let $S := S_K$ and $M := S_K/I(K)$. Consider the local cohomology $H^i_{\mathfrak{m}}(M)$, which is a $\mathbb{Z}^{V(K)}$-graded module. Now we have equality

$$\begin{aligned}\operatorname{reg}(M) &= \max\{i+j \mid H^i_{\mathfrak{m}}(M)_j \neq 0,\ j \in \mathbb{Z}\}\\ &= \max\{i+\sum_v \alpha_v \mid H^i_{\mathfrak{m}}(M)_\alpha \neq 0,\ \alpha \in \mathbb{Z}^{V(K)}\}\\ &= \max\{i-|F| \,:\, \widetilde{H}_{i-|F|-1}(\operatorname{lk}_K F) \neq 0,\ F \in K\},\end{aligned}$$

where the first equality is the local cohomology reformulation, the second follows by un-collapsing the total $\mathbb{Z}$-grading to the $\mathbb{Z}^{V(K)}$-grading, and the last equality follows from Hochster's formula [HH11, Theorem A.7.3]. Note that $\widetilde{H}_{i-|F|-1}(\operatorname{lk}_K F) \neq 0$ implies that $i-|F|-1 \geq -1$, and so $i-|F|$ is always non-negative, and we get the stated formula. □

Proof of Theorem 4.8. This follows from

$$\begin{aligned}\operatorname{pdim}_A I_X &= \max\{k \geq 0 \mid \widetilde{H}_{k-1}(\Delta^{>\underline{Z}}) \neq 0 \text{ for some } Z \in \mathcal{J}(P)\}\\ &= \max\{k \geq 0 \mid \widetilde{H}_{k-1}(\operatorname{lk}_{\mathcal{S}_X}(F)) \neq 0 \text{ for some } F \in \mathcal{S}_X\}\\ &= \operatorname{reg} \Bbbk[\mathcal{S}_X],\end{aligned}$$

where the first equality follows from Corollary 4.6 and the last equality follows from Lemma 4.13. For the middle equality, we get $\leq$ from Lemma 4.10, whereas $\geq$ follows from Lemma 4.11. □

5. Multiplicity in low degrees

We now investigate further properties of the minimal projective resolution constructed above. For the minimal resolution of I_X constructed in Theorem 4.4, equation (4.2.3) gives

$$\left[\mathcal{P}_{\min,k} : P_Z\right] = \widetilde{\beta}_{Z,k} = \dim_{\Bbbk} \widetilde{H}_{k-1}(\Delta^{>\underline{Z}}).$$

In the following subsections, we see some possible behavior exhibit by these multiplicities.

5.1. Multiplicity-free in degree zero and one.

In degrees zero and one, the topology of $\Delta^{>\underline{Z}}$ is sufficiently restricted to give the following multiplicity-free results.

Proposition 5.1. *The projective modules $\mathcal{P}_{\min,0}$ and $\mathcal{P}_{\min,1}$ are multiplicity-free.*

Proof. For $k=0$, the multiplicity of P_Z is $\dim_{\Bbbk} \widetilde{H}_{-1}(\Delta^{>\underline{Z}})$, which is 1 when $\Delta^{>\underline{Z}} = \emptyset$ and is 0 otherwise.

Now consider $k=1$. We have

$$\widetilde{\beta}_{Z,1} = \dim_{\Bbbk} \widetilde{H}_0(\Delta^{>\underline{Z}}) = \#\{\text{connected components of the 1-skeleton of } \Delta^{>\underline{Z}}\} - 1.$$

The 1-skeleton of $\Delta^{>\underline{Z}}$ is a graph whose elements are $\underline{Y} \in \mathcal{D}_X^{>\underline{Z}}$, and two vertices $\underline{Y}, \underline{Y'}$ are adjacent precisely when they are comparable. It then suffices to show that the 1-skeleton of $\Delta^{>\underline{Z}}$ has at most two connected components. Note that if $\mathcal{D}_X^{>\underline{Z}} = \emptyset$, then the 1-skeleton of $\Delta^{>\underline{Z}}$ is just the point $\emptyset$ and $\widetilde{H}_0(\Delta^{>\underline{Z}}) = 0$.

Every vertex $\underline{Y} \in \mathcal{D}_X^{>\underline{Z}}$ falls into one of the three cases (a) $Y \subsetneq Z$, (b) $Z \subsetneq Y$, or (c) Y and Z are incomparable. In case (a), if $Y, Y' \subsetneq Z$, then $Y \cap Y'$ is an order ideal. Lemma 2.11(5) gives $Z \setminus Y, Z \setminus Y' \subseteq X$. Similarly applying Lemma 2.11(5) to $Y, Y \cap Y'$ and to $Y', Y \cap Y'$ yields

$$\underline{Y} \subseteq \underline{Y \cap Y'}, \qquad \underline{Y'} \subseteq \underline{Y \cap Y'}, \qquad \underline{Z} \subsetneq \underline{Y \cap Y'}.$$

Thus all vertices in case (a) lie in one connected component.

For case (b), given $Z \subsetneq Y, Y'$, then $Y \cup Y'$ is an order ideal. By Lemma 2.11(5), both $Y \setminus Z$ and $Y' \setminus Z$ are disjoint from X. Applying the same item to $Y, Y \cup Y'$ and to $Y', Y \cup Y'$ yields

$$\underline{Y} \subseteq \underline{Y \cup Y'}, \qquad \underline{Y'} \subseteq \underline{Y \cup Y'}, \qquad \underline{Z} \subsetneq \underline{Y \cup Y'}.$$

Thus all vertices in case (b) lie in one connected component.

Finally, suppose that Y and Z are incomparable. Since $\underline{Z} \subsetneq \underline{Y}$, Lemma 2.11(6) gives

$$\underline{Y \cap Z} = \underline{Z} \sqcup (Z \setminus Y), \qquad \underline{Y \cup Z} = \underline{Z} \sqcup (Y \setminus Z).$$

Both are vertices of the 1-skeleton and are comparable with $\underline{Y}$; the first corresponds to an order ideal properly contained in Z, whereas the second corresponds to one properly containing Z. Consequently, any Y incomparable with Z belongs to one of the two components described. □

5.2. **Higher multiplicities via independence complexes.** In contrast to the behavior in degree 0 and 1, we demonstrate that multiplicities can be arbitrary large in higher degrees already when P is a height-two poset.

Let $B = (X \sqcup Y, E)$ be a (finite) bipartite graph so we can regard $E \subseteq X \times Y$. Specifying such a B is equivalent to specifying a height-two poset, namely, the corresponding poset is given by $(P_B = X \sqcup Y, \leq)$ with

$$x < y \quad \Leftrightarrow \quad \{x, y\} \in E \qquad (x \in X,\ y \in Y),$$

and no other strict relations. Conversely, the Hasse diagram of a height two poset $(P = X \sqcup Y, \leq)$, where X is of height 0 and Y is of height 1, is precisely the bipartite graph B with $P_B = P$.

A subset $\sigma \subseteq V(B)$ is *independent* if no two vertices of σ are joined by an edge of B. The *independence complex* $\operatorname{Ind}(B)$ is the simplicial complex on $V(B)$ whose faces are the independent subsets of B.

Proposition 5.2. *For a bipartite graph $B = (X \sqcup Y, E)$, the poset $\mathcal{F}\big(\operatorname{Ind}(B)\big)$ is the same as $\mathcal{D}_X$ associated to the height-two poset P_B. In particular, we have $\operatorname{sd}\operatorname{Ind}(B) = \Delta(\mathcal{D}_X \setminus \{\emptyset\}) = \Delta^{>\underline{X}}$, and hence*

$$[\mathcal{P}_{\min,k} : P_X] = \beta_{X,k} = \widetilde{H}_{k-1}(\Delta^{>\underline{X}}) \cong \widetilde{H}_{k-1}(\operatorname{Ind}(B)) \text{ for all } k \geq 0.$$

Proof. Fix $T \in \mathcal{J}(P_B)$. We show that $\underline{T}$ is an independent set of B. Indeed, note that $\underline{T} = (X \setminus T) \sqcup (T \cap Y)$, so if, on the contrary, $x \in X \setminus T$ and $y \in T \cap Y$ were adjacent in B, then $x < y$ in P_B. But T being an order ideal implies that $x \in T$, a contradiction.

Conversely, let U be an independent set of B, we claim that $T_U := \big(X \setminus (U \cap X)\big) \cup (U \cap Y) \in \mathcal{J}(P_B)$. Indeed, if $y \in U \cap y$ and $x < y$ in P_B, then independence gives $x \notin U$, so $x \in T_U$. Since $T_U = X \triangle U$, Lemma 2.11(2) gives

$$\underline{T_U} = X \triangle (X \triangle U) = U.$$

Hence we get that $\mathcal{F}(\operatorname{Ind}(B))$ are the same collection of subsets as $\mathcal{D}_X$. Since both posets use inclusion order, they are the same poset.

The last assertion follows from $\underline{X} = \emptyset$, which holds by Lemma 2.11(1). □

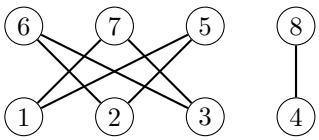
By Proposition 5.2, a bipartite graph B with $\operatorname{Ind}(B)$ that has reduced homology of dimension greater than one gives a non-multiplicity-free term we desired.

Example 5.3. Consider a bipartition B of $C_6 \sqcup K_2$ given by

$$X = \{1, 2, 3, 4\}, \qquad Y = \{5, 6, 7, 8\},$$

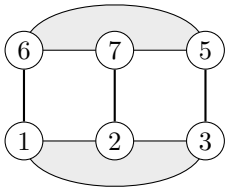
so that P_B is of the form:

P_B:



We can draw $\operatorname{Ind}(C_6)$ as follows, where the 2-simplices are filled triangles $\{1, 2, 3\}$ and $\{5, 6, 7\}$, and three additional maximal faces $\{1, 6\}, \{2, 7\}, \{3, 5\}$.

$\operatorname{Ind}(C_6)$:



Contracting the two 2-simplices yields a graph with two vertices joined by three edges, which means that we have homotopic equivalent space $|\operatorname{Ind}(C_6)| \simeq S^1 \vee S^1$, where $\vee$ here denotes the wedge sum. Also, we have $|\operatorname{Ind}(K_2)| = S^0$.

Note that an independent set of a disjoint union $G \sqcup H$ is the union of an independent set of G and one of H, i.e. $\operatorname{Ind}(G \sqcup H) = \operatorname{Ind}(G) * \operatorname{Ind}(H)$, where $*$ denotes the simplicial join (faces are the union of those in the two components). Since joining with S^0 is homotopic to taking the suspension $\Sigma(-)$, and suspension preserves wedge sums, we get a homotopy equivalence

$$\begin{aligned} |\operatorname{Ind}(B)| &= |\operatorname{Ind}(C_6) * \operatorname{Ind}(K_2)| \\ &\simeq (S^1 \vee S^1) * S^0 \simeq \Sigma(S^1 \vee S^1) \simeq S^2 \vee S^2. \end{aligned}$$

Hence, for $k \geq 0$, we have

$$\widetilde{H}_k(\operatorname{Ind}(B)) \cong \widetilde{H}_k(S^2 \vee S^2) \cong \widetilde{H}_k(S^2)^{\oplus 2} = \begin{cases} \Bbbk^{\oplus 2}, & k = 2, \\ 0, & \text{otherwise.} \end{cases}$$

In particular, we have $[\mathcal{P}_{\min,k+1} : P_X] = 2$, i.e. $\mathcal{P}_{\min,3}$ is not multiplicity-free.

In general, we can wedge more spheres to get arbitrary multiplicity in reduced homology.

Example 5.4. For $m \geq 1$ and $k \geq 2$, take a finite triangulation of $K = \bigvee_{j=1}^{m} S^{k-1}$. Then $\widetilde{H}_{k-1}(K) \cong \Bbbk^m$.

The final ingredient we need is the following realization theorem.

Theorem 5.5. ([NR09, Proposition 6.2]) *For every finite simplicial complex K, there exists a finite bipartite graph B_K such that* $\operatorname{Ind}(B_K) \simeq \Sigma K$.

Corollary 5.6. *For every $k \geq 2$ and every $m \geq 1$, there exist a finite height-two poset P and an order ideal $X \in \mathcal{J}(P)$ such that the multiplicity $[\mathcal{P}_{\min,k} : P_X] = m$ in the degree k term of any minimal projective resolution $\mathcal{P}_{\min,\bullet}$ of the $\Sigma_{\mathcal{J}(P)}$-module I_X.*

Proof. Take $K = \bigvee_{j=1}^{m} S^{k-1}$, then we get that $\dim_{\Bbbk} \widetilde{H}_{k-1}(K) = m$ as seen in Example 5.4. Let $B_K = X \sqcup Y$ be the bipartite graph guaranteed by Theorem 5.5 and take $P = P_{B_K}$, then the claim follows from Proposition 5.2. □

Appendix A. A simplicial construction for thin modules

The projective resolution in Section 3 is an instance of the simplicial replacement (bar construction of [Rie14, Definition 4.2.1]) of a diagram of projective modules. This explains why it looks very much like an economoical reformulation of bar construction. For some modules, this construction always gives a complex of projectives, although the complex need not be exact.

Let $A = \Bbbk Q/I$ be a basic finite-dimensional bound-quiver algebra, and let M be a *thin* right A-module, i.e. $\dim_{\Bbbk} Me_x \leq 1$ for every $i \in Q_0$. Let Q_M be the subquiver of Q whose vertices are

$$(Q_M)_0 := \{x \in Q_0 \mid Me_x \neq 0\}$$

and whose arrows are those $a : x \to y$ for which right multiplication induces a non-zero map $Me_x \to Me_y$. The key assumption we need is that non-zero vectors $m_x \in Me_x$ can be chosen so that

$$m_x a = m_y \qquad \text{for every arrow } a : x \to y \text{ in } Q_M. \tag{A.0.1}$$

The quiver Q_M is acyclic. Indeed, if it contained an oriented cycle, then (A.0.1) would imply that every positive power of this cycle acts non-trivially on one of the vectors m_x. This contradicts the admissibility of I, since every sufficiently long path belongs to I.

For every $x \in (Q_M)_0$, let $\pi_x \colon P_x = e_x A \to M$ be the homomorphism determined by $\pi_x(e_x) = m_x$. Since each non-zero space Me_x is spanned by m_x, these maps give a surjection

$$d_0 := (\pi_x)_{x \in (Q_M)_0} \colon \bigoplus_{x \in (Q_M)_0} P_x \to M. \tag{A.0.2}$$

Consider the small category $\mathcal{C}_M$ associated to M given as follows. We take $(Q_M)_0$ as the objects of $\mathcal{C}_M$. For $x, y \in (Q_M)_0$, define

$$\mathcal{C}_M(y, x) := \{\mu_p : P_y \to P_x \mid p : x \to y \text{ is a path on } Q_M\},$$

where $\mu_p : P_y \to P_x$ is the homomorphism of projective A-modules given by left-multiplying p. In particular, two paths $p, p' : x \to y$ determine the same morphism of $\mathcal{C}_M$ precisely when

$\mu_p = \mu_{p'}$, and the identity of x is $\mu_{e_x} = \mathrm{id}_{P_x}$. If there is no path between x, y on Q_M, then we take $\mathcal{C}_M(y,x) = \emptyset$. Composition is given by $\mu_p\mu_q = \mu_{pq}$ whenever $p : x \to y$ and $q : y \to z$.

Define a functor

$$F_M \colon \mathcal{C}_M \longrightarrow \operatorname{proj} A, \qquad i \longmapsto P_i, \qquad \mu_p \longmapsto \mu_p.$$

By (A.0.1), it follows that $\pi_x\mu_p = \pi_y$, so the homomorphisms π_x form a cocone over F_M with vertex M, which means that there is a surjection induced by d_0 of (A.0.2)

$$\operatorname{colim}_{\mathcal{C}_M} F_M \cong \Big(\bigoplus_{x\in\mathcal{C}_M} P_x\Big)\Big/ \sum_{\substack{p:x\to y\\ m\in P_y}} (m - pm)A \twoheadrightarrow M.$$

For $k \geq 0$, let $N_k(\mathcal{C}_M)$ be the set of composable strings of k non-identity morphisms

$$x_* = \big(x_0 \xrightarrow{f_1} x_1 \xrightarrow{f_2} \cdots \xrightarrow{f_k} x_k\big).$$

In particular, $N_0(\mathcal{C}_M)$ is the object set of $\mathcal{C}_M$. For $0 \leq r \leq k$, write $x_*^{\widehat{r}}$ for the string obtained by deleting i_r and (only in the case of $r \neq 0, k$) composing the adjacent morphisms. Define

$$\mathcal{P}_k(M) := \bigoplus_{x_*\in N_k(\mathcal{C}_M)} F_M(i_0)^{(x_*)} = \bigoplus_{x_*\in N_k(\mathcal{C}_M)} P_{i_0}^{(x_*)}.$$

Here $F_M(i_0)^{(x_*)}$ denotes a labelled copy of $F_M(i_0) = P_{i_0}$. For $k > 0$, define

$$d_k \colon \mathcal{P}_k(M) \longrightarrow \mathcal{P}_{k-1}(M), \qquad u^{(x_*)} \mapsto \big(F_M(f_1)(u)\big)^{(x_*^{\widehat{0}})} + \sum_{r=1}^{k} (-1)^r u^{(x_*^{\widehat{r}})}.$$

Thus the zeroth face changes the initial projective through $F_M(f_1)$, while every other face acts as the identity on the initial projective.

Proposition A.1. *For a thin A-module M, the maps d_k above yield a complex of A-modules*

$$\cdots \longrightarrow \mathcal{P}_2(M) \xrightarrow{d_2} \mathcal{P}_1(M) \xrightarrow{d_1} \mathcal{P}_0(M) \xrightarrow{d_0} M \to 0,$$

which is exact at $\mathcal{P}_0(M)$ when M is faithful.

Proof. For a string x_* and $0 \leq r \leq k$, let $\varepsilon_r^{x_*}$ be the homomorphism defines analogous to (3.1.3). The zeroth face is induced by $F_M(f_1)$, while every other face is the identity map between the corresponding labelled copies of $F_M(i_0)$. Functoriality of F_M gives the simplicial identities

$$\varepsilon_i^{x_*^{\widehat{j}}}\varepsilon_j^{x_*} = \varepsilon_{j-1}^{x_*^{\widehat{i}}}\varepsilon_i^{x_*} \qquad (0 \leq i < j \leq k).$$

The only identity involving two non-identity homomorphisms is the one obtained by deleting x_0 and x_1, and it follows from $F_M(f_2)F_M(f_1) = F_M(f_2f_1)$. For $0 \leq i < j \leq k$, the equal composites $\varepsilon_i^{x_*^{\widehat{j}}}\varepsilon_j^{x_*}$ and $\varepsilon_{j-1}^{x_*^{\widehat{i}}}\varepsilon_i^{x_*}$ occur in $d_{k-1}d_k$ with coefficients $(-1)^{i+j}$ and $(-1)^{i+j-1}$, respectively. Hence $d_{k-1}d_k = 0$ for $k \geq 2$.

For a morphism $f : x_0 \to x_1$ and $u \in P_{i_0}$, the the equality $\pi_{x_1} \circ F_M(f) = \pi_{x_0}$ gives

$$d_0d_1\big(x^{(f)}\big) = \pi_{x_1}F_M(f)(u) - \pi_{x_0}(u) = 0.$$

Hence the displayed sequence is a complex.

Suppose now that M is faithful. Then $Q_M = Q$, since every vertex idempotent and every arrow acts non-trivially on M. Let $\iota_x : P_x \hookrightarrow \mathcal{P}_0(M)$ be the canonical inclusion. Let $\overline{d}_0 : \operatorname{coker} d_1 \to M$ be the map induced by d_0. For every non-trivial path $p : x \to y$, we have

$$d_1\big(e_y^{(\mu_p)}\big) = \iota_x(p) - \iota_y(e_y).$$

Since path classes span each e_xAe_y, it follows that $(\operatorname{coker} d_1)e_y$ is spanned by $[\iota_y(e_y)]$. This class maps to $m_y \neq 0$, which spans Me_y. Hence the restriction $\overline{d}_0|_{(\operatorname{coker} d_1)e_y} : (\operatorname{coker} d_1)e_y \to Me_y$ is an isomorphism for every $y \in Q_0$. Therefore $\overline{d}_0$ is an isomorphism, and $\ker d_0 = \operatorname{im} d_1$. □

Example A.2. For $A = \Sigma_{\mathcal{J}(P)}$ and $M = I_X$, choose $m_Y \in I_Xe_Y$ to be the functional dual to b_{YX}. Then $\mathcal{C}_{I_X}$ can be identified with the poset category $\mathcal{D}_X$: there is a non-identity morphism $Z \to T$ precisely when $\underline{Z} < \underline{T}$, and this morphism is represented by the left-multiplication map $\mu_{T,Z} : P_Z \to P_T$ given by $a \mapsto b_{TZ}a$. The set $N_k(\mathcal{C}_M)$ becomes the k-chains in $\Delta(\mathcal{D}_X)$.

Example A.3. For a simple module S_i, the category $\mathcal{C}_{S_i}$ has only the object i and its identity morphism. Hence $\mathcal{P}_0(S_i) = P_i$ and $\mathcal{P}_k(S_i) = 0$ for $k > 0$, so the construction reduces to the projective cover $P_i \to S_i$. It is a projective resolution only when S_i is itself projective. This shows a non-resolution $\mathcal{P}_\bullet$.

Now we relate $\mathcal{P}_\bullet(M)$ to the two-sided simplicial bar construction used in [Rie14, Definition 4.2.1]; we note that this does not really give an alternative proof of Proposition A.1 – the proof we shown here is specialized case of the reason exactly the reason why the simplicial bar construction yields a complex. We summarise the translation of setting as follows.

Bar construction setting	Our $\mathcal{C}_M$ setting
$\mathcal{D}$	$\mathcal{C}_M$
$\mathcal{M}$	$s(\operatorname{Mod} A)$, the category of simplicial right A-modules.
$F : \mathcal{D} \to \mathcal{M}$	The composite $c \circ F_M$, where $F_M : \mathcal{C}_M \to \operatorname{proj} A$ and c regards an ordinary module as a constant simplicial module.
$G : \mathcal{D}^{\mathrm{op}} \to \mathrm{sSet}$	The constant functor $G = *$ with value in $* \in \mathrm{sSet}$.
$\vec{d} : [k] \to \mathcal{D}$	A string $x_* = (x_0 \xrightarrow{f_1} \cdots \xrightarrow{f_k} x_k)$, but allowing identity morphisms.
$Gd_k \otimes Fd_0$	$* \otimes cF_M(x_0) \cong cF_M(x_0)$;[1]
$B_k(G, \mathcal{D}, F)$	The constant simplicial module associated to $\bigoplus_{x_*} F_M(i_0)$.
Face maps	Homomorphisms $\varepsilon_r^{x_*}$, whose alternating sum gives d_k.
Degeneracy maps	The homomorphisms inserting identity morphisms into x_*.

Let us explain the translation of $\mathcal{M}$ and F now. Recall that a simplicial right A-module is a functor $\Delta^{\mathrm{op}} \to \operatorname{Mod} A$, where Δ is the category of finite non-empty totally ordered sets $[n] = \{0 < \ldots < n\}$ and order-preserving maps; roughly speaking, it is a sequence $\{M_k\}_{k \geq 0}$ of right A-modules (where *degree k term* M_k is the functor evaluated at $[k]$) equipped with homomorphisms that delete or repeat positions, satisfying compatibility identities. Every right A-module N determines a *constant simplicial module* cN, with $(cN)_k = N$ for all $k \geq 0$ where all structure maps are id_N. We take

$$\mathcal{M} = s(\operatorname{Mod} A), \qquad F = c \circ F_M : \mathcal{C}_M \longrightarrow s(\operatorname{Mod} A),$$

where the inclusion $\operatorname{proj} A \hookrightarrow \operatorname{Mod} A$ is understood. Thus, we have $F(x) = cP_x$ for each $x \in \mathcal{C}_M = Q_M$.

Next, we unfold the definition of the simplicial bar construction to see how (the modules underlying) $\mathcal{P}_\bullet$ shows up in it. Writing out the evaluation of a string $\vec{d} : [k] \to \mathcal{D}$ yields a sequence $d_0 \xrightarrow{f_1} d_1 \xrightarrow{f_2} \cdots \xrightarrow{f_k} d_k$. These d_j's are our $x_j \in Q_M$, and so with the chosen F, G we have a summand

$$Gd_k \otimes Fd_0 = * \otimes c(F_M(x_0)) \cong cF_M(x_0)$$

indexed by $\vec{d}$ of $B_k(G, \mathcal{D}, F)$. Note that $*$ here is the terminal object in the category sSet of simplicial sets, that is the singleton set at every degree with all structure map being identity; taking the simplicial tensor with $*$ is naturally isomorphic to applying the identity functor, which explains the last isomorphism.

Next, we want to clarify the differential of the simplicial bar complex. For each fixed k, we have $k+1$ face maps $\partial_k^{(r)} : B_k(G, \mathcal{D}, F) \to B_{k-1}(G, \mathcal{D}, F)$ for $0 \leq r \leq k$. As we show in the table above, these are our ε_r maps. To see this, for $r = k$, its restriction to $Gd_k \otimes Fd_0$ is given by $G(f_k) \otimes \mathrm{id}_{F(d_0)}$, which translates to $\epsilon_k^{(x_*)} : p^{(x_*)} \mapsto p^{(\widehat{x_*^k})}$ in our setting. For $r = 0$, the restriction to the same summand is

$$\mathrm{id}_{G(d_k)} \otimes F(f_1) : G(d_k) \otimes F(d_0) \longrightarrow G(d_k) \otimes F(d_1),$$

which is $\epsilon_0^{(x_*)} : p^{(x_*)} \mapsto (F_M(f_1)(p))^{(x_*^{\hat{0}})}$ in our setting. For $0 < r < n$, $\partial^{(r)}$ composes f_r and f_{r+1}, and leaves the element of $F(d_0)$ unchanged, which means that the same term-omitting map

[1] The $\otimes$ on the left is the simplicial tensoring, not tensor product of modules.

$\epsilon_r^{(x_*)} : p^{(x_*)} \mapsto p^{(x_*^{\widehat{r}})}$ in our setting. The differential of the (not-yet-normalized) simplicial bar complex $B_\bullet = B_\bullet(G, \mathcal{D}, F)$ is

$$d_k^{\mathrm{un}} : B_k \longrightarrow B_{k-1}, \qquad d_k^{\mathrm{un}} = \sum_{r=0}^{k} (-1)^r \partial_k^{(r)}.$$

Note that $d_{k-1}^{\mathrm{un}} d_k^{\mathrm{un}} = 0$ is just a consequence of $\partial_{k-1}^{(r)} \partial_k^{(s)} = \partial_{k-1}^{(s-1)} \partial_k^{(r)}$ for any $0 \le r < s \le k$; by the same argument we use in Proposition A.1.

To normalise the bar complex, we need to consider, for each $0 \le r \le k-1$, the degeneracy map

$$s_{k-1}^{(r)} : B_{k-1} \longrightarrow B_k,$$

which inserts the identity morphism id_{i_r} into the string at the object i_r. They satisfy

$$\partial_k^{(r)} s_{k-1}^{(q)} = \begin{cases} s_{k-2}^{(q-1)} \partial_{k-1}^{(r)}, & r < q, \\ \mathrm{id}_{B_{k-1}}, & r = q \text{ or } r = q+1, \\ s_{k-2}^{(q)} \partial_{k-1}^{(r-1)}, & r > q+1. \end{cases} \tag{A.0.3}$$

Define

$$C_k = \sum_{r=0}^{k-1} \operatorname{im} s_{k-1}^{(r)} \subseteq B_k, \text{ and } C_0 = 0.$$

Then (A.0.3) implies that $d_k^{\mathrm{un}}(C_k) \subseteq C_{k-1}$. Consequently, we have a quotient complex $\overline{B}_\bullet$, called the *normalized bar complex* given by

$$\overline{B}_k := B_k / C_k, \text{ with differential } d_k(b + C_k) = d_k^{\mathrm{un}}(b) + C_{k-1}.$$

Since $\mathcal{C}_M$ has no non-identity isomorphisms, we have

$$\overline{B}_k \cong \bigoplus_{\substack{x_0 \xrightarrow{f_1} \cdots \xrightarrow{f_k} x_k \\ f_r \neq \mathrm{id} \text{ for all } r}} F_M(x_0) = \bigoplus_{f_* = (x_0 \to \cdots \to x_k) \in N_k(\mathcal{C}_M)} F_M(x_0).$$

Moreover, the differential is the alternating sum of the maps $\partial_k^{(r)}$ after deleting every term that contains an identity morphism, which implies that $\overline{B}_\bullet \cong \mathcal{P}_\bullet(M)$.

As a consequence of the identification with bar construction, we get that

$$H_0\big(\mathcal{P}_\bullet(M)\big) \cong \operatorname{colim}_{\mathcal{C}_M} F_M, \qquad H_k\big(\mathcal{P}_\bullet(M)\big) \cong L_k \operatorname{colim}_{\mathcal{C}_M} F_M \quad (k > 0),$$

where L_k denotes the k-th left-derived functor. In the general setting of $\mathcal{C}_M$, these higher derived colimits may not vanish and the colimit may not be the same M itself, whereas Theorem 3.3 proves that, in the case when $\mathcal{C}_M = \mathcal{D}_X$ (Example A.2), this will not be the case.

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